
DATA INTEGRITY

Definition Reliability of a set of data.

Explanation It is the overall state of data being error-free and usable, maintained through accuracy and consistency throughout the data lifecycle.

Scenario A medical clinic must ensure that a patient's allergy information remains unchanged and accessible when transferred from the intake form to the electronic health record.

Question (3-4 marks) Explain why data integrity is essential for a business when making strategic decisions.

Mark Scheme:

- High-quality data ensures decisions are based on facts rather than errors.
- Reliable data prevents financial loss caused by acting on incorrect information.
- Consistent data allows for accurate comparisons over different time periods or departments.

ARCHIVE

Definition Transfer of less frequently used data to a backing store.

Explanation Archiving frees up primary storage space and improves system performance by moving historical data to secondary storage while keeping it accessible if needed.

Scenario A bank moves ten-year-old transaction records from its high-speed servers to a secure offline storage facility to reduce database maintenance costs.

Question (3 marks) Describe the difference between a data backup and an archive.

Mark Scheme:

- A backup is a copy of data for restoration after loss, whereas an archive is the original data moved to long-term storage.

- Backups are updated frequently; archives contain historical data no longer in active use.
 - Archives are often kept for legal or regulatory compliance.
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DATA GOVERNANCE

Definition Systems and procedures for ensuring data is reliable and secure.

Explanation Governance provides the administrative framework and policies that dictate how an organization manages its data assets to maintain high quality and security.

Scenario A global corporation implements a policy that defines who is responsible for updating client addresses and how those changes must be verified.

Question (4 marks) Explain the role of data governance in maintaining data integrity within a large organization.

Mark Scheme:

- Establishes clear ownership and accountability for data sets.
 - Sets standardized procedures for data entry to ensure consistency.
 - Implements security protocols to prevent unauthorized modification.
 - Provides the rules for audit trails to track changes over time.
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VALIDATION

Definition Techniques and processes designed to ensure data is reasonable and follows specific rules.

Explanation Validation acts as a filter at the point of entry to catch obvious errors, though it cannot guarantee the data is 100% correct.

Scenario An e-commerce website uses a rule to ensure that a customer cannot enter a negative number in the "quantity" box during checkout.

Question (3 marks) "Validation ensures that data is completely accurate." Discuss this statement.

Mark Scheme:

- Validation only checks if data is reasonable or follows set rules, not if it is true.
 - A user could enter a wrong but "reasonable" value (e.g., a wrong but valid email address) which passes validation.
 - Accuracy requires data to reflect actual real-world values, which validation cannot always confirm.
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ACCURACY

Definition Data must be correct and free from errors/inconsistencies; it must reflect actual values.

Explanation Accuracy is the degree to which stored data matches the real-world facts it represents, such as a correct phone number.

Scenario A delivery company loses revenue because a customer's address was entered with the wrong house number, despite the format being correct.

Question (3 marks) Identify two ways a company can improve the accuracy of its stored data.

Mark Scheme:

- Use of double-entry or verification checks during input.
 - Regular data cleansing and auditing to remove outdated or incorrect records.
 - Providing users with the ability to review and confirm their data (e.g., a "Confirm Details" screen).
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CONSISTENCY

Definition Data must follow set rules and formats.

Explanation Consistency ensures that identical data points across different tables or systems are stored in the same way to avoid conflicting information.

Scenario A retail chain ensures all its branches record dates in the DD/MM/YYYY format to prevent errors during regional sales analysis.

Question (3 marks) Explain how a lack of data consistency can negatively affect a multi-national business.

Mark Scheme:

- Different units of measurement (e.g., miles vs. km) lead to incorrect combined reports.
 - Conflicting data across systems causes confusion for management during decision-making.
 - Difficulties in merging databases during company acquisitions or migrations.
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COMPLETENESS

Definition Inclusion of all necessary data points.

Explanation Completeness requires that every field necessary for a process or decision is populated, as missing data can lead to skewed results.

Scenario A marketing survey is deemed useless because 40% of the respondents left the "Age" and "Location" fields blank.

Question (3 marks) Analyze the impact of incomplete data on a business's predictive analytics.

Mark Scheme:

- Missing variables lead to biased or incorrect pattern identification.
 - Predictive models will produce unreliable forecasts, leading to poor resource allocation.
 - Decreases the "Value" of the big data collected.
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RELIABILITY

Definition Users must be able to trust the data.

Explanation Reliability is the user's confidence in the system's output, achieved when data is consistently available and demonstrably accurate.

Scenario A stockbroker relies on a real-time data feed that has zero downtime and has been verified for accuracy over the last five years.

Question (3 marks) Identify three components that contribute to the reliability of a dataset.

Mark Scheme:

- Robust security protocols to prevent tampering.
 - Regular data backups to ensure availability after a failure.
 - Rigorous validation rules to maintain data standards.
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SECURITY

Definition Measures used to prevent unauthorised access, modification, or deletion.

Explanation Security protects the integrity of data by safeguarding it from corruption, tampering, or malicious breaches.

Scenario A law firm uses AES-256 encryption and biometric scanners to ensure that only authorized partners can view sensitive case files.

Question (4 marks) Explain how security measures support data integrity.

Mark Scheme:

- Access controls prevent unauthorized users from deleting or modifying records.
 - Encryption ensures that even if data is stolen, it cannot be read or altered without the key.
 - Prevents malicious corruption of data (e.g., via malware or hackers).
 - Safeguards the system from accidental unauthorized changes by employees.
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DATA BACKUP

Definition Regularly creating copies of data to protect it from loss or corruption for restoration after system errors.

Explanation Backups provide a "safety net," allowing a business to return to a known good state if the primary data is lost due to hardware failure or a cyberattack.

Scenario A school's IT department runs a full system backup every Friday night and incremental backups every other evening at 9 PM.

Question (3 marks) Describe why a business should store its backups in an off-site location.

Mark Scheme:

- Protects the data copy from local physical disasters like fire or flooding.
 - Ensures a copy is available if the entire local network is compromised by ransomware.
 - Facilitates disaster recovery planning and business continuity.
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DATA ACCESS CONTROLS

Definition Limits on who can access/modify data to prevent unauthorised changes.

Explanation Controls ensure that users only have the permissions necessary for their specific job roles, following the principle of least privilege.

Scenario A hospital receptionist can view patient appointment times but is blocked from accessing clinical surgery notes.

Question (4 marks) Discuss the benefits of implementing role-based access controls (RBAC) in a database.

Mark Scheme:

- Reduces the risk of accidental data deletion by untrained staff.
 - Enhances data privacy by restricting sensitive info to authorized personnel only.
 - Simplifies management as permissions are granted to roles, not individuals.
 - Provides a clear structure for auditing which group performed specific actions.
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AUDIT TRAILS

Definition Tracking any changes made to data, by whom, and when.

Explanation Audit trails provide a chronological record of system activities, allowing administrators to investigate errors or security breaches.

Scenario After discovering an error in the payroll database, the manager checks the logs to see which username modified the salary field on Tuesday morning.

Question (4 marks) Identify four specific details that should be recorded in a comprehensive audit trail.

Mark Scheme:

- The timestamp (date and time) of the action.
 - The username of the person who performed the action.
 - The type of action performed (e.g., create, delete, modify).
 - The specific data element or record that was affected.
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SECURE BY DESIGN

Definition Fundamental principle where programmers consider integrity and implementation during the planning stage of IT systems.

Explanation This proactive approach ensures that security and integrity are built into the foundation of a system rather than added as an afterthought.

Scenario A software company holds a mandatory "integrity planning" meeting before a single line of code is written for their new banking app.

Question (3 marks) Explain why the "Secure by Design" approach is more effective than adding security patches later.

Mark Scheme:

- Identifies and mitigates architectural vulnerabilities before they are exploited.
 - Reduces the complexity and cost of fixing security flaws in a finished system.
 - Ensures data integrity is considered as a core functional requirement.
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DATA DICTIONARY

Definition A blueprint or plan for a database or dataset.

Explanation It serves as a centralized documentation tool that defines the structure, data types, and rules for every field in a database.

Scenario Before building a student registration database, the developer creates a document specifying that "Student_ID" must be a 6-digit integer.

Question (3 marks) Identify three components found in a standard data dictionary.

Mark Scheme:

- Table name and Field name.
 - Data type (e.g., INT, VARCHAR).
 - Validation rules or constraints (e.g., primary key).
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METADATA

Definition Data or information that describes other data to provide deeper meaning or understanding.

Explanation Metadata helps systems and users organize, find, and use data effectively by providing context, such as the date a photo was taken.

Scenario A digital library uses tags such as "Author," "Publication Date," and "Genre" to help users search through its collection of PDFs.

Question (3 marks) Explain how metadata is used by a database management system (DBMS).

Mark Scheme:

- Used by the query optimizer to find data more efficiently.
 - Enforces data types and lengths specified in the data dictionary.
 - Facilitates the categorization of unstructured or semi-structured data.
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SQL DATA TYPES

Definition Indicates the type of operations allowed on a data field (e.g., integer for math, string for text).

Explanation Selecting the correct data type is critical for storage efficiency, performance, and ensuring that valid operations can be performed on the data.

SQL DATA TYPES LIST (Shortened for brevity as per tutor style):

- **INT**: Whole numbers (usually -2,147,483,648 to 2,147,483,647).
- **SMALLINT**: Smaller integers (-32,768 to 32,767).
- **BIGINT**: Very large integers.
- **TINYINT**: Small numbers (0 to 255).
- **DECIMAL**: Precise decimal numbers (e.g., 10,2).
- **FLOAT/REAL**: Approximate floating-point numbers.
- **DATE**: YYYY-MM-DD.
- **TIME**: HH:MM:SS.
- **DATETIME/TIMESTAMP**: Combined date and time for tracking.
- **CHAR**: Fixed-length text string.
- **VARCHAR**: Variable-length string; space-efficient.
- **TEXT**: For large blocks of text.
- **NCHAR/NVARCHAR**: Unicode compatible for multilingual text.
- **BINARY/VARBINARY**: For files like images/audio.
- **BOOLEAN**: TRUE/FALSE.
- **JSON**: Format for passing data between systems.

Question (4 marks) A database designer needs to store a person's age (0-120) and their exact bank balance. Recommend and justify the most suitable SQL data types.

Mark Scheme:

- **TINYINT** for age because it covers the 0-255 range and is memory-efficient.
- **DECIMAL** for bank balance because it provides exact precision required for financial data.
- Using **FLOAT** for money is incorrect as it is an "approximate" value.

DATA VALIDATION (PROCESS)

Definition Implementing rules to attempt to reduce the chance of errors in stored data.

Explanation It consists of three phases: User Input, the Check (system compares input to rules), and the Post-check Action (passing data or triggering an error).

Scenario A flight booking system checks if the departure date is in the future before allowing the user to proceed to payment.

Question (4 marks) Describe the steps a system takes when a validation check fails.

Mark Scheme:

- The system prevents the record from being saved to the database.
 - An error message is output to the user explaining why the data was rejected.
 - The logic requires the user to re-enter the data.
 - The system loops back to the "Check" phase once new data is input.
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PRESENCE CHECK

Definition Ensures mandatory fields are not blank.

Explanation This check prevents the database from containing "null" or missing values in critical fields, supporting the principle of completeness.

Scenario A signup form for a newsletter will not allow the "Submit" button to be clicked unless the "Email Address" field contains text.

Question (3 marks) Provide the pseudocode logic for a presence check on a variable called `user_input`.

Mark Scheme:

- `IF LENGTH(user_input) == 0:`
 - `RETURN False`
 - `ELSE: RETURN True`
-

RANGE CHECK

Definition Ensures data falls between two accepted values.

Explanation Range checks prevent unrealistic numerical or date data from being entered by setting upper and lower boundaries.

Scenario An employee timesheet system rejects any entry where the "Hours Worked" is less than 0 or greater than 168 hours per week.

Question (3 marks) Explain how a range check contributes to data integrity in a school's grading system.

Mark Scheme:

- Prevents impossible scores (e.g., 110%) being entered for a test marked out of 100.
 - Ensures data accuracy by limiting entries to a realistic "reasonable" scope.
 - Reduces the need for manual data cleaning later.
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LOOKUP CHECK

Definition Checks input against a pre-existing list/table stored outside the data table.

Explanation Lookup checks ensure that the data entered matches an existing record in a separate related table, often used for IDs or codes.

Scenario A warehouse system checks an entered "Part_Number" against the master inventory list to ensure the item actually exists.

Question (3 marks) Explain why a lookup check is better than a simple list check for a database of 5,000 product codes.

Mark Scheme:

- A list check is only suitable for a small number of predefined entries.
 - Lookups reference external tables which are easier to update centrally than hard-coded lists.
 - Searching a 5,000-item table is more efficient and prevents human error in entry.
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LIST CHECK

Definition Compares input to a small list of predefined acceptable entries.

Explanation List checks are often implemented as dropdown menus or radio buttons to restrict user choices to valid options only.

Scenario A cinema booking app forces the user to select their ticket type from "Child," "Adult," or "Senior."

Question (3 marks) Identify two advantages of using a list check in a user interface.

Mark Scheme:

- Speeds up data entry as users do not have to type the values.
 - Guarantees data consistency as users cannot use different spellings (e.g., "Child" vs. "Kid").
 - Eliminates the possibility of "rogue" data entries.
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LENGTH CHECK

Definition Checks that character count meets minimum/maximum requirements.

Explanation This check ensures that strings like passwords or ID numbers contain the correct number of characters to be valid.

Scenario A bank requires all online banking passwords to be at least 12 characters long to improve security.

Question (3 marks) Provide the pseudocode logic for a length check that requires a `user_input` to be exactly 8 characters.

Mark Scheme:

- `IF LENGTH(user_input) != 8:` (or equivalent `>7 AND <9` logic)
 - `RETURN False`
 - `ELSE: RETURN True`
-

FORMAT CHECK

Definition Ensures data follows a specific pattern.

Explanation Format checks (or picture checks) use templates to ensure data like postcodes or ID numbers are structured correctly (e.g., LLNN NLL).

Scenario A vehicle registration system ensures that every license plate entered begins with two letters followed by two numbers.

Question (3 marks) Explain why a format check is used for entering email addresses.

Mark Scheme:

- Ensures the entry contains necessary symbols like '@' and '.'.
 - Prevents users from accidentally entering their name or phone number in the email field.
 - Improves consistency for automated mailing systems.
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CHECK DIGIT

Definition A number added to a string to detect errors (incorrect entry/transposition).

Explanation The system recalculates the digit using an algorithm; if the result matches the provided digit, the entry is likely accurate.

Scenario A barcode scanner at a supermarket calculates a final digit to ensure the product code wasn't misread due to a smudge on the label.

Question (4 marks) Describe how a check digit detects a "transposition error."

Mark Scheme:

- A transposition error occurs when two adjacent numbers are swapped during entry.
 - The algorithm used to calculate the check digit (e.g., weighted sums) is sensitive to the position of numbers.
 - Swapping numbers changes the final calculation result.
 - The system's calculated digit will no longer match the stored check digit, triggering an error.
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DATA REDUNDANCY

Definition Unnecessary replication of data in multiple locations within a database.

Explanation Redundancy leads to inefficiencies and errors, although "intentional duplication" is sometimes used for backups.

Scenario A company stores a customer's full home address in the "Orders" table, the "Invoices" table, and the "Marketing" table separately.

Question (4 marks) Discuss the problems caused by high levels of data redundancy.

Mark Scheme:

- Increased storage costs as the database grows exponentially.
 - Data inconsistency; updating an address in one table but not others leads to conflicting records.
 - Maintenance implications; managing and updating redundant data wastes time and effort.
 - Slower system performance when running queries across bloated tables.
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NORMALISATION

Definition The process of breaking down complex data tables into successively smaller/simpler tables to remove redundancy and ensure related data is properly linked.

Explanation Normalisation is a multi-step design process (1NF to 3NF) that optimizes database structure for integrity and efficiency.

Scenario A developer takes a giant spreadsheet of sales data and splits it into three linked tables: Customers, Products, and Orders.

Question (4 marks) Explain the benefits of a fully normalised database.

Mark Scheme:

- Minimizes data redundancy, reducing storage requirements.
 - Prevents update anomalies (inconsistencies) since data is only stored in one place.
 - Improves data integrity through the use of primary and foreign keys.
 - Makes the database more flexible and easier to maintain or scale.
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FIRST NORMAL FORM (1NF)

Definition A table state where each cell contains atomic values, unique column names, same data type in columns, and a primary key is used.

Explanation 1NF is the first step in normalisation, focusing on removing "repeating groups" or multiple items in a single cell.

Scenario In a library database, a single "Books_Borrowed" cell containing "The Hobbit, 1984, Dracula" is split into three separate rows to meet 1NF.

Question (3 marks) Identify the four rules required for a table to be in First Normal Form.

Mark Scheme:

- Each cell must contain single, indivisible (atomic) values.
 - All entries in a column must be of the same data type.
 - Each column must have a unique name.
 - A primary key must be used to identify each record.
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SECOND NORMAL FORM (2NF)

Definition A table state that satisfies 1NF and ensures there are no partial dependencies.

Explanation 2NF applies to tables with composite primary keys; it ensures every non-key column depends on the *entire* primary key, not just part of it.

Scenario A "Project_Assignments" table is split so that "Employee_Name" (which only depends on Employee_ID) is moved to a separate "Employees" table.

Question (3 marks) Explain the concept of a "partial dependency."

Mark Scheme:

- Occurs in tables with a composite primary key (made of two or more columns).
 - Happens when a non-key attribute depends on only one part of the primary key.
 - These attributes must be moved to their own separate tables to satisfy 2NF.
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THIRD NORMAL FORM (3NF)

Definition A table state that satisfies 2NF and ensures transitive dependencies have been removed.

Explanation In 3NF, "non-key" columns cannot depend on other "non-key" columns; all data must depend *only* on the primary key.

Scenario In an "Orders" table, the "Supplier_Phone" column is moved to a separate "Suppliers" table because it depends on "Supplier_ID," not the "Order_ID."

Question (3 marks) State the rule for a table to move from 2NF to 3NF.

Mark Scheme:

- The requirements for 2NF must be satisfied.
 - Transitive dependencies must be removed.
 - Non-key attributes that depend on other non-key attributes are moved to their own table.
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PRIMARY KEY

Definition A code/number/name that uniquely identifies a single record in a database.

Explanation The primary key is the "unique ID" that ensures no two rows in a table are identical, preventing confusion between records.

Scenario A university assigns every student a unique 8-digit "Student_Number" to ensure records aren't mixed up for students with the same name.

Question (3 marks) Identify three characteristics of a good primary key.

Mark Scheme:

- It must be unique for every record (no duplicates).
 - It cannot contain "null" or empty values.
 - It should be concise and meaningful (e.g., an ID number rather than a full paragraph).
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FOREIGN KEY

Definition A column/columns in a table that links to unique data (primary key) in another table.

Explanation Foreign keys create the "relationships" between tables, allowing the DBMS to cross-reference data while keeping tables simple.

Scenario In an "Orders" table, the "Customer_ID" column acts as a link to the master "Customers" table where the full name and address are stored.

Question (3 marks) Explain how foreign keys support referential integrity.

Mark Scheme:

- Ensures that a record in a child table must correspond to a valid record in the parent table.
 - Prevents "orphaned" records (e.g., an order for a customer who doesn't exist).
 - Enforces constraints that prevent the deletion of a primary key if it is still being used as a foreign key elsewhere.
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RECORD

Definition An individual row in a data table.

Explanation A record represents a single complete unit of information, such as one specific customer or one transaction.

Scenario In a car dealership database, row #452 contains the make, model, year, and price for a specific 2022 Ford Focus.

Question (3 marks) Describe the relationship between fields and records.

Mark Scheme:

- Fields are the vertical columns representing specific categories/attributes.
 - A record is the horizontal row containing a collection of fields for one item.
 - Multiple fields combined together form one complete record.
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ATOMIC DATA

Definition Data representing a single attribute that cannot be broken down further.

Explanation Ensuring data is atomic is the core requirement of 1NF; it prevents "data stuffing" where multiple pieces of info are in one cell.

Scenario Instead of a single "Address" field, a database uses "House_Number," "Street," "City," and "Postcode" to ensure the data is fully searchable.

Question (3 marks) Explain why the field "Full_Name" (e.g., "Mr. John Smith") is not considered atomic data.

Mark Scheme:

- It can be broken down into smaller components: Title, First Name, and Surname.
 - Non-atomic data makes sorting (e.g., by Surname) much more difficult.
 - Violates the first rule of Normalisation (1NF).
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LOGICAL DATA MODEL

Definition A plan for how data elements are structured and relate to each other, often represented as an entity-relationship diagram (ERD).

Explanation The logical model is the conceptual map of the database, created before any actual tables are built in the software.

Scenario An architect for a new library database draws a diagram showing how "Books" connect to "Authors" and "Borrowers."

Question (4 marks) List four components that should be included in a Logical Data Model.

Mark Scheme:

- Entities (which become tables).
 - Attributes (which become fields).
 - Relationships (e.g., one-to-many).
 - Unique identifiers or Primary Keys.
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ENTITY

Definition Real-world objects/concepts (e.g., customers, products).

Explanation In a database design, an entity is the "thing" you are storing information about, and it eventually becomes a table.

Scenario In a zoo database, "Animal," "Enclosure," and "Keeper" are identified as the three main entities.

Question (3 marks) Describe the relationship between an Entity and a Table.

Mark Scheme:

- An entity is the conceptual object/idea in the design phase.
 - A table is the physical implementation of that entity in the database software.
 - Every entity must be normalised before it becomes a table.
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ATTRIBUTE

Definition Characteristics/properties describing entities (e.g., customer_name).

Explanation Attributes provide the specific details for an entity and eventually become the columns (fields) of a table.

Scenario For the entity "Smartphone," the attributes include "Screen_Size," "Battery_Capacity," and "Model_Name."

Question (3 marks) State two requirements for attributes in a logical data model.

Mark Scheme:

- Every attribute must be atomic (indivisible).
 - Every attribute must have a unique and meaningful name within its entity.
 - They must accurately describe a property of the parent entity.
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RELATIONSHIP

Definition Defines how entities connect (one-to-one, one-to-many, many-to-many).

Explanation Relationships describe the business logic of how data interacts, such as "one customer can place many orders".

Scenario A music app defines a "one-to-many" relationship between an "Artist" and their "Albums."

Question (3 marks) Describe the notation used to show a "one-to-many" relationship in an ERD.

Mark Scheme:

- Shown using a line connecting two entities.
- The "one" side is often shown with a single line.
- The "many" side is denoted by a three-pronged line (crow's foot) or the letter 'n'.

DBMS (DATABASE MANAGEMENT SYSTEM)

Definition A system that provides tools for creating and manipulating databases.

Explanation The DBMS is the software layer (like MySQL or Oracle) that sits between the users and the raw data, managing security and access.

Scenario A hospital uses a DBMS to ensure that doctors can update patient files while simultaneously preventing unauthorized hackers from accessing the server.

Question (4 marks) Identify four tasks performed by a DBMS to maintain a database.

Mark Scheme:

- Providing tools for creating tables and defining schemas.
- Authentication and authorization of users (security).
- Enforcing referential integrity and validation rules.
- Managing data backups and recovery processes.

BIG DATA

Definition Very large and complex data sets generated by computer systems that are difficult to manage using traditional data-processing tools.

Explanation Big data is characterized by the "5 Vs" and requires specialized infrastructure like distributed file systems or NoSQL databases to process.

Scenario A weather satellite network generates petabytes of data every hour from millions of sensors worldwide.

Question (4 marks) Explain why traditional relational databases often struggle to manage big data.

Mark Scheme:

- The "Volume" is too high for single-server storage.
- The "Velocity" (speed of generation) exceeds the processing power of traditional systems.
- Relational databases require "Structured" data, but big data often comes in "Unstructured" formats like video or text.
- Latency issues arise when performing complex calculations on massive datasets.

VOLUME (5 VS)

Definition The sheer amount of data constantly generated.

Explanation Volume refers to the massive scale of data, measured in terabytes or petabytes, often coming from sources like social media or IoT devices.

Scenario Facebook must manage over 4 million 'likes' every single minute.

Question (3 marks) Explain the storage challenges associated with high data volume.

Mark Scheme:

- Requires massive physical space and high hardware costs.
 - Needs scalability, often provided by cloud storage like AWS or Azure.
 - Requires distributed file systems to store data across multiple nodes.
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VELOCITY (5 VS)

Definition The speed at which data is generated.

Explanation Velocity is the rate of data inflow; high velocity requires systems that can process and analyze data in real-time to be useful.

Scenario Twitter handles approximately 360,000 posts every minute, requiring immediate processing for trending topics.

Question (3 marks) Describe one social benefit of high-velocity data processing.

Mark Scheme:

- Enables real-time monitoring for emergency services or traffic management.
 - Allows for immediate fraud detection in banking transactions.
 - Supports dynamic scheduling for public transportation based on current demand.
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VARIETY (5 VS)

Definition Range of formats and structures.

Explanation Variety covers the different "shapes" of data: Structured (tables), Unstructured (video/images), and Semi-structured (JSON/XML).

Scenario A travel agency collects data in the form of Excel spreadsheets, Instagram photos, and customer feedback emails.

Question (4 marks) Distinguish between structured and unstructured data.

Mark Scheme:

- Structured data is well-organized and fits easily into traditional relational tables.
 - Unstructured data does not follow a set logical model (e.g., raw text files or video).
 - Structured data is easier to query/analyze using SQL.
 - Unstructured data requires specialized "NoSQL" databases or metadata for processing.
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VERACITY (5 VS)

Definition Quality/accuracy of the data.

Explanation Veracity addresses the uncertainty of data; if data is biased, unregulated, or incorrect, it has low veracity and leads to poor decisions.

Scenario An AI model for health advice performs poorly because it was trained on unverified and conflicting comments from public social media forums.

Question (3 marks) Explain why veracity is a major concern when using social media as a data source.

Mark Scheme:

- Social media data is often unregulated and can contain intentional misinformation.
 - It may be highly biased toward specific demographics, skewing analysis results.
 - High uncertainty makes it difficult for businesses to trust the "Value" of the insights.
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VALUE (5 VS)

Definition Potential for data to bring significant rewards.

Explanation Value is the ultimate goal of big data; it is the actionable intelligence that improves decision-making, profits, or customer satisfaction.

Scenario A grocery store analyzes purchase patterns to realize they can increase sales by 15% if they move the bread aisle next to the milk.

Question (3 marks) Analyze the relationship between Veracity and Value.

Mark Scheme:

- If data has low veracity (low quality), it has little to no value for decision-making.
 - High-quality, accurate data (high veracity) is required to unlock significant rewards/profits.
 - Mined information must be "actionable" to truly have value.
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INFRASTRUCTURE

Definition Hardware and software resources supporting data and processes.

Explanation Infrastructure includes the physical sensors, cloud servers, and network cables that allow an organization to collect, store, and transmit data.

Scenario A city's smart-lighting project requires thousands of sensors, a high-speed fiber-optic network, and a central cloud server.

Question (3 marks) Identify the three main groups of infrastructure/services for big data.

Mark Scheme:

- Collection (e.g., IoT devices, social media).
 - Storage (e.g., Cloud providers, data warehouses).
 - Transmission (e.g., Bandwidth, fiber optics).
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IOT (INTERNET OF THINGS)

Definition Interconnection of everyday objects through the internet.

Explanation IoT turns "dumb" objects into data-collection tools by equipping them with sensors and connectivity, allowing for automation and remote monitoring.

Scenario A smart refrigerator monitors milk levels and automatically adds a carton to the user's online grocery shopping list when it runs low.

Question (3 marks) Discuss one ethical concern related to the infrastructure of IoT devices.

Mark Scheme:

- Manufacturing requires rare earth metals often extracted from conflict zones.
 - Constant power needs for millions of devices contribute to high global energy consumption.
 - Rapid obsolescence leads to massive amounts of "e-waste" containing toxic substances.
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BANDWIDTH

Definition Amount of data transmitted over a connection in a given time.

Explanation High bandwidth is essential for big data transmission to avoid bottlenecks and ensure that massive datasets can move between systems quickly.

Scenario A video streaming service upgrades its cables to 10Gbps fiber optics to support 4K streaming for thousands of simultaneous users.

Question (3 marks) Explain how low bandwidth affects the performance of a distributed system.

Mark Scheme:

- Increases latency (delay) between requesting and receiving data.
 - Creates bottlenecks that slow down synchronization across data nodes.
 - Prevents real-time processing of high-velocity data.
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DATA SILO

Definition Data stored across different departments/systems; integration is complex.

Explanation Silos occur when data is "isolated," preventing a unified view of the organization and leading to inconsistency or missed insights.

Scenario The sales department and the customer service department use two different databases that cannot talk to each other, so agents can't see a customer's recent orders.

Question (3 marks) Explain how a data warehouse helps to eliminate data silos.

Mark Scheme:

- It acts as a central store for raw data collected from varied sources.
 - Provides a "unified view" of the entire organization's data.
 - Simplifies integration by processing data into a standardized format.
-

LATENCY

Definition The delay between requesting and receiving data.

Explanation Latency is a critical metric in transmission; high latency can render real-time systems like self-driving cars or stock trading platforms dangerous or useless.

Scenario A gamer experiences a 2-second delay between pressing a button and their character moving because the game server is located in another country.

Question (3 marks) Identify two factors that contribute to high latency in big data transmission.

Mark Scheme:

- Physical distance of the storage node from the user.
 - Overloaded networks with insufficient bandwidth.
 - High processing time required for complex calculations on large volumes of data.
-

DATA MINING

Definition Extracting meaningful information using statistical techniques (clustering, classification, anomaly detection).

Explanation Mining uses machine learning and AI to find hidden patterns in raw big data that are not obvious to human analysts.

Scenario A credit card company uses an algorithm to flag a transaction in a foreign country as "potentially fraudulent" based on the user's past behavior.

Question (4 marks) Identify four strands of data mining techniques used in business.

Mark Scheme:

- Clustering (grouping similar items).
 - Classification (labeling data).
 - Anomaly detection (finding outliers).
 - Regression (predicting continuous outcomes).
-

DATA WAREHOUSING

Definition Collection, storage, and management of raw data in a centralised repository for a unified view.

Explanation A warehouse serves as the "holding tank" for big data, optimized specifically for analytics rather than day-to-day transactions.

Scenario A global retailer aggregates sales data from 50 countries into one master repository to analyze yearly performance.

Question (3 marks) Compare a standard relational database with a data warehouse.

Mark Scheme:

- Databases are used for daily operational tasks (EPOS); warehouses are for long-term analytics.
 - Databases hold current data; warehouses hold massive amounts of historical raw data.
 - Warehouses are optimized for complex queries across many different data sources.
-

DESCRIPTIVE ANALYTICS

Definition Summarising historical data to understand "what happened".

Explanation This is the simplest form of analytics, using charts and reports to visualize past performance, such as last month's total revenue.

Scenario A manager prints a "Monthly Attendance Report" to see how many employees were late in October.

Question (3 marks) State two tools used to present descriptive analytics to management.

Mark Scheme:

- Dashboards with real-time charts and graphs.
 - Aggregated reports summarizing key performance indicators (KPIs).
 - Simulations or models showing past trends.
 -
-

PREDICTIVE ANALYTICS

Definition Analysing historical patterns to predict future outcomes (causes/effects).

Explanation Predictive analytics uses the "what happened" data to build models that forecast "what might happen next".

Scenario A toy company analyzes 10 years of sales data to predict that "space-themed" toys will be the top sellers next Christmas.

Question (3 marks) Explain how predictive analytics can improve urban planning.

Mark Scheme:

- Predicts future traffic congestion patterns to plan new roads.
 - Forecasts population growth to determine where to build new schools.
 - Helps in predictive maintenance of systems (e.g., water pipes) before they fail.
-

PRESCRIPTIVE ANALYTICS

Definition Suggesting actions required to optimise results based on predictions.

Explanation The most advanced analytics level; it doesn't just predict the future, it recommends exactly what the business *should do* to achieve the best result.

Scenario An airline's system suggests increasing ticket prices by 5% and adding an extra flight to Paris next Tuesday to maximize profit based on a predicted surge in demand.

Question (4 marks) Distinguish between predictive and prescriptive analytics.

Mark Scheme:

- Predictive analytics focuses on future outcomes and probabilities.
 - Prescriptive analytics suggests specific "actionable" decisions to optimize those outcomes.
 - Predictive: "Sales will go up in winter" vs. Prescriptive: "Order 500 extra coats now".
 - Prescriptive models require more complex simulations and AI to weight different choices.
-

HOW TO ANSWER GUIDE

1. DEFINE (1-2 marks)

- **What to include:** Use the "textbook" definition. Always include the purpose or a key rule (e.g., for 1NF, mention "atomic data").
- **What to avoid:** Don't use the word you are defining in the answer (e.g., "Data integrity is the integrity of data" = 0 marks).
- **Pro-tip:** For data types, always state the range or what it specifically stores (e.g., "TINYINT is 0 to 255").

2. EXPLAIN (3-4 marks)

- **Structure:** Use the **P-E-I** (Point, Evidence/Example, Impact) structure.
- **Keywords:** Use terms like "Consequently," "Therefore," or "This leads to..." to show a logical chain of reasoning.
- **Focus:** Explain *how* it maintains integrity or *why* it is used in a database context.

3. DISCUSS/EVALUATE (6-8 marks)

- **Structure:** You must provide a **balanced argument**. Start with the benefits, then the limitations/drawbacks, and end with a conclusion.
- **Scenario Use:** The examiner wants to see you apply the theory. If the question mentions a "Hospital," talk about "Patient records," not just "records."
- **What the examiner wants:** Analysis of the *impact*. Don't just list facts; explain how the technology changes the way the organization operates.

4. COMMON MISTAKES (TOPIC 12)

- **Integrity vs. Security:** They are not the same! Security *supports* integrity by preventing tampering, but integrity also requires accuracy and consistency.
- **Validation vs. Accuracy:** Validation only checks if the data is *reasonable*. It cannot tell if a user lied about their birthdate as long as the date is in a valid format.
- **Normalisation Rules:** Many students confuse 2NF and 3NF. Remember: 2NF = No partial dependencies (look for composite keys); 3NF = No transitive dependencies (non-keys depending on non-keys).
- **Big Data "Vs":** Don't just list them. Be prepared to explain how they interact (e.g., how high *Velocity* makes maintaining *Veracity* harder).

VIRTUALISATION

Definition The act of creating a virtual version of something, such as a computer system, server, storage device, or network resource, which behaves like the original.

Explanation Virtualisation allows multiple simulated environments to run on a single physical hardware platform, improving efficiency and testing capabilities.

Scenario A software development firm uses a single powerful server to simulate several different gaming consoles to test their new title's performance across various platforms.

Question (3 marks) Explain one reason why virtualisation is beneficial for a company that develops mobile applications.

Mark Scheme:

- It allows developers to mimic different target systems (e.g., various Android versions) without buying physical hardware.
- This reduces the cost of maintaining a physical library of every hardware variation.
- It enables rapid testing of software in environments that behave exactly like the intended original device.

DESKTOP VIRTUALISATION

Definition Running multiple operating systems on a single desktop computer.

Explanation It allows a user to access different OS environments (like Windows on a Mac) simultaneously on one physical machine.

Scenario An IT consultant uses a MacBook but runs a virtualised version of Windows 11 to use a specific legacy accounting software that is not compatible with macOS.

Question (3 marks) Describe how desktop virtualisation can improve a worker's productivity.

Mark Scheme:

- Users can access software exclusive to different operating systems without switching physical computers.
 - It allows for a "thin client" approach where resources are stored on a remote server, reducing local maintenance.
 - It enables a consistent working environment that can be accessed from various physical locations.
-

STORAGE VIRTUALISATION

Definition Pooling multiple physical storage devices into a single virtual storage pool, simplifying management as a single system.

Explanation It abstracts physical disks into a single logical volume, making it easier for administrators to allocate space without worrying about which specific drive the data is on.

Scenario A large corporation connects fifty separate hard drives across different servers and presents them to the network as one giant 500TB "Virtual Drive".

Question (3 marks) Identify two advantages of implementing storage virtualisation in a data centre.

Mark Scheme:

- Simplifies management by allowing multiple devices to be controlled as a single system.
 - Improves resource utilisation by pooling unused space from various physical disks.
 - Increases flexibility, as administrators can add or remove physical storage without disrupting the virtual pool.
-

NETWORK VIRTUALISATION

Definition Creating virtual networks independent of the underlying physical infrastructure, allowing for network isolation.

Explanation It allows administrators to split a single physical network into multiple secure, independent logical networks.

Scenario A university uses one physical cable layout but creates separate virtual networks for "Student Guest Wi-Fi" and "Staff Financial Records" to ensure the groups cannot interact.

Question (4 marks) Explain how network virtualisation supports the security of an organisation.

Mark Scheme:

- It provides network isolation, keeping sensitive administrative resources separate from general user traffic.
 - Malware or breaches in one virtual network are restricted from spreading to others (sandbox effect).
 - Administrators can apply different security protocols and access controls to each virtual network independently.
 - It allows for the creation of secure testing environments that do not affect the live physical infrastructure.
-

RESOURCE UTILISATION

Definition Maximizes use of owned physical resources; running tools/services from different ecosystems on one platform improves efficiency.

Explanation Virtualisation ensures that expensive hardware does not sit idle by allowing one server to perform the work of many.

Scenario A business replaces five under-utilised physical servers with one powerful server running five virtual instances, saving on electricity and floor space.

Question (3 marks) Explain how virtualisation leads to better resource utilisation.

Mark Scheme:

- Most physical servers only use a small percentage of their capacity; virtualisation allows multiple OSs to share that spare capacity.
 - It reduces "hardware sprawl" by consolidating services onto fewer physical machines.
 - This leads to increased efficiency in terms of power consumption and physical maintenance.
-

ISOLATION

Definition Separation between the virtual system and the host platform; issues like malware are restricted to the virtual instance.

Explanation Isolation creates a "sandbox" where any crashes or security threats within a virtual machine do not affect the main physical system or other virtual machines.

Scenario A cybersecurity researcher opens a suspicious email attachment inside a virtual machine so that if it contains a virus, it cannot infect the actual computer's hard drive.

Question (3 marks) Describe the benefit of isolation when testing new, unstable software.

Mark Scheme:

- If the unstable software causes a system crash, only the virtual instance is affected, not the host machine.
- It allows for "sandboxing," which prevents software bugs from corrupting files in the underlying physical ecosystem.
- The researcher can simply delete the corrupted virtual instance and start over without needing to reinstall the host OS.

FLEXIBILITY AND SCALABILITY

Definition Virtual environments are easier to adapt to changing workloads and requirements compared to physical hardware.

Explanation Virtual machines can be created, duplicated, or given more RAM/CPU power instantly via software, whereas physical hardware requires manual upgrades.

Scenario During a massive 24-hour flash sale, an e-commerce site quickly spins up ten additional virtual servers to handle the surge in traffic, then deletes them once the sale ends.

Question (3 marks) Distinguish between the scalability of physical servers versus virtual servers.

Mark Scheme:

- Physical scalability requires purchasing, delivering, and manually installing new hardware components.
 - Virtual scalability is near-instant, allowing resources to be adjusted through a management interface.
 - Virtual systems allow for "on-demand" growth, whereas physical systems are often over-provisioned and sit idle.
-

DISASTER RECOVERY

Definition Facilitates creation of snapshots and backups.

Explanation Because virtual machines are stored as simple files, they can be backed up and restored to different hardware much faster than physical systems.

Scenario After a flood destroys the server room, a company uses its virtual server backups to restart its entire IT department on a cloud provider's hardware in under an hour.

Question (4 marks) Explain why virtualisation is considered a vital component of a disaster recovery plan.

Mark Scheme:

- Virtual machines are hardware-independent, meaning they can be restored to any physical machine that has a hypervisor.
 - It allows for the creation of snapshots, which provide an immediate "rollback" point if data is corrupted.
 - Backing up an entire virtual disk file is simpler and more reliable than backing up individual files and configurations manually.
 - It reduces the Recovery Time Objective (RTO) as systems do not need a full manual OS reinstallation.
-

SNAPSHOT

Definition A saved copy of a system, including all data and settings, at a given point in time.

Explanation A snapshot acts like a "save point" in a video game, allowing a system to be instantly reverted if a future change goes wrong.

Scenario An IT manager takes a snapshot of the company's database server at 9:00 AM before applying a risky security patch at 9:05 AM.

Question (3 marks) Describe the mechanism of using a snapshot if a software update fails.

Mark Scheme:

- The snapshot captures the exact state of the data and settings before the update.
 - If the update causes a failure, the administrator can trigger a "restore" or "rollback".
 - This immediately returns the system to the 100% working state captured in the snapshot.
-

COMPLEXITY

Definition Setting up and managing virtual environments requires specialist software and skilled personnel; maintaining isolation and security is more complex than on host environments.

Explanation While it simplifies hardware, it adds a "software layer" that requires expert knowledge to configure and secure correctly.

Scenario A small business struggles to implement virtualisation because their only IT staff member is not trained in managing hypervisors or virtual network switches.

Question (3 marks) Explain why security can be more complex in a virtualised environment.

Mark Scheme:

- Administrators must secure both the host physical machine and every individual virtual machine.
 - If the hypervisor (the program managing the VMs) is compromised, all virtual systems are at risk.
 - Maintaining proper isolation between virtual networks requires advanced configuration knowledge.
-

COST

Definition While it can save money long-term, initial setup (specialist software, powerful hardware) and additional IT staff training can be expensive.

Explanation Virtualisation shifts spending from many cheap physical units to a few very expensive, high-performance units and licensing fees.

Scenario A school saves \$5,000 a year on electricity by using virtualisation, but they had to spend \$20,000 upfront on a high-end server and hypervisor licenses.

Question (3 marks) Justify the high initial cost of virtualisation for a growing company.

Mark Scheme:

- Long-term savings are achieved through reduced energy consumption and cooling costs.
- It reduces the need to purchase new physical hardware every time a new service is required.
- Increased uptime and faster disaster recovery prevent financial losses from system downtime.

PERFORMANCE

Definition Virtual solutions share hardware resources, potentially leading to bottlenecks if too many Virtual Machines run or if hardware is underpowered/misconfigured.

Explanation Because multiple systems are fighting for the same CPU and RAM, a virtual machine will always be slightly slower than the same machine running directly on physical hardware.

Scenario A video editing VM runs slowly because three other virtual machines on the same server are currently performing heavy database calculations.

Question (3 marks) Explain how "resource contention" can affect the performance of a virtual machine.

Mark Scheme:

- Multiple VMs share the same physical CPU, RAM, and disk storage.
- If one VM uses an excessive amount of resources, it leaves less for the others, causing lag.
- This creates "bottlenecks" where the virtual system must wait for the physical hardware to become available.

VIRTUAL MACHINES (VMs)

Definition Simulate an entire computer system, providing access to a complete operating system.

Explanation A VM includes its own kernel and OS, making it heavy but providing the strongest possible isolation between applications.

Scenario A developer runs a virtualised version of Linux Ubuntu on their Windows laptop to test if their code works in a non-Windows environment.

Question (4 marks) Identify two characteristics of Virtual Machines and explain one drawback compared to containerisation.

Mark Scheme:

- Characteristics: They simulate full hardware virtualisation and include a complete Guest OS.
- Drawback: VMs consume significantly more resources (CPU/RAM) because each one must run its own full operating system.
- Drawback: They have much larger image sizes and take longer to start up compared to containers.

CONTAINERISATION

Definition Focuses on a narrower scope, providing access to a specific function or application.

Explanation Containers share the host machine's operating system kernel, making them lightweight, fast, and highly portable.

Scenario A web developer bundles a single "Login Function" into a container so it can be moved from a testing laptop to a live cloud server without any changes.

Question (4 marks) Discuss why containerisation is preferred over Virtual Machines for cloud-native applications.

Mark Scheme:

- Containers have minimal overhead and faster startup times, which is essential for high-speed cloud services.
 - They are highly portable across different environments since they share the host OS kernel.
 - They allow for "microservices," where different parts of an app are updated independently.
 - However, they offer lower isolation and higher security risk than VMs if the shared kernel is compromised.
-

HYPERVISOR

Definition A program that runs and manages virtual machines.

Explanation The hypervisor is the "manager" software that sits between the physical hardware and the virtual machines, allocating resources to each VM as needed.

Scenario An administrator uses VMware ESXi as the hypervisor to divide a server's 128GB of RAM among ten different virtual machines.

Question (3 marks) Describe the role of the hypervisor in hardware abstraction.

Mark Scheme:

- It hides the physical hardware details from the virtual machines.
 - It presents "virtual" versions of the CPU, RAM, and Storage to the Guest OS.
 - It manages the scheduling of tasks to ensure multiple VMs can share the same physical processor efficiently.
-

ENCAPSULATION

Definition To group or bundle together related data and/or functions.

Explanation In virtualisation, encapsulation allows an entire system (OS, settings, and files) to be stored as a single file, making it easy to move or copy.

Scenario A technician copies a single 40GB .vmdk file to a USB drive to move an entire virtualised server from the London office to the Paris office.

Question (3 marks) Explain how encapsulation benefits the portability of virtualised systems.

Mark Scheme:

- It bundles the entire operating system, applications, and data into a single file.
 - This file can be moved or copied to any other machine running the same hypervisor.
 - It eliminates the need to reinstall software or reconfigure settings when migrating to new hardware.
-

DISTRIBUTED SYSTEMS

Definition A collection of independent computing nodes that work together to perform a common goal.

Explanation Instead of one central computer doing everything, the work is spread across many different machines in different locations to improve speed and reliability.

Scenario Google uses a distributed system to process search queries, using thousands of servers worldwide to find an answer in milliseconds.

Question (4 marks) Explain two benefits of using a distributed system for a global video streaming service.

Mark Scheme:

- **Performance:** Using Content Delivery Networks (CDNs) allows users to access video from a local server, reducing lag and response time.
 - **Failure Resistance:** If one server node fails, the other nodes in the network can continue to provide the service.
 - **Scalability:** New nodes can be added to the system easily to handle more users without replacing the entire infrastructure.
 - **Data Replication:** Movies can be mirrored across multiple sites to ensure they are always available.
-

NODE

Definition A key point in a network relating to a specific connection or device.

Explanation In a distributed system, each node is an independent computer that has its own memory and processor but communicates with others to solve a problem.

Scenario In a blockchain network, every participant's computer acts as a node that stores a copy of the transaction ledger.

Question (3 marks) Describe the relationship between nodes and the overall goal of a distributed system.

Mark Scheme:

- Each node is an independent computing unit with its own resources.
 - Nodes are distributed across different locations and platforms.
 - They communicate and coordinate with each other to perform a common task as if they were a single system.
-

FAILURE RESISTANCE

Definition High fault tolerance; if one node fails or is cyber-attacked, others continue to operate and restore the system.

Explanation Distributed systems avoid "single points of failure" by ensuring that no one machine is essential for the whole system to function.

Scenario Even though a major data centre in Virginia goes offline due to a power cut, Amazon's website stays online because other nodes in Europe and Asia take over the workload.

Question (3 marks) Explain how distributed systems achieve high failure resistance.

Mark Scheme:

- They eliminate single points of failure by spreading tasks across multiple nodes.
 - Data is replicated across different sites, so a backup is always available.
 - If one node is attacked or fails, the remaining nodes detect the failure and continue operating.
-

CONCURRENCY

Definition Multiple tasks/operations occur simultaneously on different nodes; essential for high performance in large organisations.

Explanation Concurrency allows a system to handle thousands of users at the same exact time by splitting the requests across many different processors.

Scenario During a busy ticket launch, 50,000 fans can all click "Buy" at the same second because the ticket system processes their requests concurrently across hundreds of nodes.

Question (3 marks) Explain why concurrency is a critical requirement for modern web services.

Mark Scheme:

- It allows the system to handle a high volume of simultaneous user interactions.
 - Spreading the workload across different nodes prevents any single machine from being overwhelmed.
 - It ensures high performance and low response times during peak usage periods.
-

REPLICATION

Definition Data is duplicated across nodes; requires systems to ensure data consistency across all instances to avoid errors.

Explanation Replication provides redundancy so that if data is lost in one location, a perfect copy exists elsewhere, though it is difficult to keep all copies identical in real-time.

Scenario A bank replicates its database in three different cities so that if a fire destroys the London branch's records, the exact same data is safe in Manchester and New York.

Question (3 marks) Identify one benefit and one challenge of data replication in a distributed system.

Mark Scheme:

- **Benefit:** Protects against data loss and improves access speed by having data closer to the user.

- **Challenge:** Maintaining consistency across all nodes so that every user sees the same information.
 - **Challenge:** Requires high bandwidth and complex software to synchronise updates across all instances.
-

LATENCY

Definition The delay between requesting and receiving data.

Explanation In distributed systems, latency is often caused by the physical distance data must travel through network cables or bottlenecks in processing power.

Scenario A user in Australia experiences high latency (a 300ms delay) when trying to access a website hosted only on a single server in London.

Question (3 marks) Explain how the physical distance of a node affects latency.

Mark Scheme:

- Data must travel through network infrastructure (cables/routers) which has a finite speed.
 - Greater distances involve more "hops" between network points, increasing the time taken.
 - High latency results in a "laggy" experience for the end user.
-

HCI (HUMAN COMPUTER INTERACTION)

Definition HCI focuses on how users control and interact with computer systems to meet their needs.

Explanation It is a multidisciplinary field that combines computer science and psychology to make technology more effective and easier for humans to use.

Scenario A software team researches how elderly users struggle with mouse clicks and decides to implement voice commands to make the system easier for them.

Question (3 marks) Describe the primary goal of HCI in software development.

Mark Scheme:

- To ensure that systems are enjoyable, easy to use, and intuitive for the user (improving User Experience).
 - To create interfaces that allow users to complete tasks with maximum efficiency and minimum errors.
 - To ensure that technology is accessible to a diverse range of users with different needs.
-

USER EXPERIENCE (UX)

Definition The extent to which users find systems enjoyable, easy, and intuitive.

Explanation UX covers the "feeling" of using a product, focusing on whether it meets the user's needs without causing frustration.

Scenario A banking app has a high UX rating because it allows users to check their balance with a single fingerprint scan rather than typing a long password.

Question (3 marks) Explain the difference between User Interface (UI) and User Experience (UX).

Mark Scheme:

- UI is the specific part of the system the user interacts with (e.g., buttons, screens).
 - UX is the overall experience or "feeling" the user has when using those components.
 - A system can have a beautiful UI but a poor UX if it is slow or difficult to navigate.
-

USER INTERFACE (UI)

Definition The part of the experience the user interacts with.

Explanation The UI includes all visual, auditory, and tactile elements (like menus, icons, and sounds) that allow a human to give commands to a machine.

Scenario The UI of a microwave consists of the physical buttons, the digital clock display, and the "beep" sound it makes when finished.

Question (3 marks) Identify three visual elements that form part of a Graphical User Interface (GUI).

Mark Scheme:

- **Layout:** The logical arrangement of elements on the screen.
 - **Typography:** The fonts used to show importance and improve readability.
 - **Icons:** Small pictures that act as shortcuts for tools or actions.
-

USABILITY

Definition Speed, responsiveness, and simplicity of processes.

Explanation Usability measures how effectively a user can achieve their goal, focusing on how quickly they can learn the system and how many mistakes they make.

Scenario A flight booking site has poor usability because it takes 15 different screens and 20 minutes just to buy a single ticket.

Question (3 marks) Identify two factors that contribute to high usability in a mobile app.

Mark Scheme:

- **Efficiency:** Minimizing the number of steps/clicks needed to complete a task.
 - **Learnability:** Providing clear instructions and an intuitive design that is easy to remember.
 - **Responsiveness:** The system reacts quickly to user input without long loading delays.
-

CONSISTENCY

Definition Similar elements/actions presented the same way across the system to help predict outcomes.

Explanation Consistency allows users to transfer their knowledge from one part of the app to another (e.g., the "Save" icon is always in the top right).

Scenario In a word processor, every menu uses the same font and colour scheme, and "Right-Click" always opens the same list of options.

Question (3 marks) Explain how consistency improves the "learnability" of a new IT system.

Mark Scheme:

- Users can predict how the system will react based on their previous actions.
 - It reduces the cognitive workload as users do not have to learn new rules for every different screen.
 - Familiar patterns (like standard icons) help users find tools faster without searching through manuals.
-

FEEDBACK

Definition Visual cues, sounds, or haptic feedback to maintain clear communication between system and user.

Explanation Feedback confirms that the system has received the user's command and is currently processing it, preventing the user from getting confused.

Scenario When a user clicks "Upload," a progress bar appears and a small "ping" sound plays once the file is successfully saved.

Question (3 marks) Describe two different types of feedback a system might provide to a user.

Mark Scheme:

- **Visual Feedback:** A spinning loading icon or a change in button colour to show a click was registered.
 - **Auditory Feedback:** A beep or notification sound to alert the user of an error or completion.
 - **Haptic Feedback:** A vibration in a smartphone or game controller to simulate a physical touch.
-

ACCESSIBILITY

Definition Design for diverse needs (color contrast, font size, screen readers).

Explanation Accessibility ensures that people with disabilities—such as visual impairment or motor issues—can use the IT system effectively.

Scenario A government website provides high-contrast mode and "Alt-text" on all images so that blind citizens can use screen-reading software to navigate the page.

Question (4 marks) Identify four features that can be implemented to improve the accessibility of a website.

Mark Scheme:

- Adjustable typeface or font size for users with low vision.
 - High-contrast colour options for users with colour blindness.
 - Screen reader compatibility (tags/metadata) to read text aloud.
 - "Alt-text" descriptions for images and captions for videos.
-

PHYSICAL ERGONOMICS

Definition Focuses on posture, movement, and physical workload to minimize musculoskeletal disorders.

Explanation It involves designing physical equipment and workspaces (like chairs and keyboards) to fit the human body and prevent long-term injury.

Scenario An office manager provides all staff with adjustable chairs and footrests to ensure their backs are straight and their feet are flat on the floor.

Question (3 marks) Explain how proper physical ergonomics can reduce health risks for office workers.

Mark Scheme:

- Adjustable furniture allows for correct sitting posture, reducing strain on the spine.
 - Curved keyboards and trackballs reduce muscle strain in the wrists/hands.
 - Proper monitor placement and lighting reduce eye strain and glare-related headaches.
-

COGNITIVE ERGONOMICS

Definition Interaction demands on a conscious level.

Explanation Cognitive ergonomics focuses on how the human brain processes information, aiming to reduce mental fatigue and distractions in an interface.

Scenario A flight control system is designed to show only the most critical warnings in red to ensure the pilot is not distracted by hundreds of unimportant green icons.

Question (3 marks) Identify three strands of cognitive ergonomics used in interface design.

Mark Scheme:

- **Workload:** Designing interfaces to minimize mental effort and fatigue.
 - **Memory:** Using familiar patterns so users don't have to remember complex steps.
 - **Attention:** Avoiding visual/auditory clutter to keep the user focused on the task.
-

MENU TYPES

Definition Graphical features providing quick access to tools.

Explanation Different menu structures (like Hamburgers or Sidebars) are chosen based on the screen size and the number of categories the user needs to see.

Scenario A mobile news app uses a "Hamburger Menu" (three horizontal lines) to hide navigation links and save precious screen space for the articles.

Question (4 marks) Recommend the most suitable menu type for:

1. A smartphone app with limited screen space.
2. A complex professional video editor with 50+ tools.

Mark Scheme:

- **Smartphone:** Hamburger menu, as it tucks links away until they are needed.
- **Video Editor:** Mega menu or Vertical sidebar, as they allow for many subcategories to be visible at once.

- **Justification:** Contextual menus (right-click) could also be used in the editor for specific tool settings.
-

ICONS

Definition Small pictures acting as shortcuts; provide consistency and accessibility.

Explanation Icons are universal symbols (like a magnifying glass for "Search") that allow users to understand functions quickly without reading text.

Scenario An international travel app uses a "House" icon for the home screen so that users who speak different languages can all find their way back easily.

Question (3 marks) Describe two benefits of using icons instead of text labels in a mobile interface.

Mark Scheme:

- Icons take up less physical screen space than long text words.
 - They are universally recognized, aiding users who speak different languages or have low literacy.
 - They provide visual consistency across different devices and operating systems.
-

W3C (WORLD WIDE WEB CONSORTIUM)

Definition World Wide Web Consortium provides accessibility guidance.

Explanation The W3C sets the international standards (like WCAG) that web developers must follow to ensure the internet is usable by everyone, regardless of disability.

Scenario A web developer checks their code against W3C guidelines to ensure that their new online shop is fully compatible with Braille displays.

Question (3 marks) State the significance of the W3C in modern web development.

Mark Scheme:

- It acts as the guiding body for web accessibility standards.
 - It ensures that websites work consistently across different browsers and devices.
 - It provides frameworks that help developers meet legal requirements for disability access.
-

CLOUD STORAGE

Definition Resources provided by a third party and accessed via the internet.

Explanation Data is stored on massive server farms owned by providers like Amazon or Google, allowing users to access their files from any device with an internet connection.

Scenario A student saves their essay to Microsoft OneDrive so they can start writing it on a school PC and finish it later on their phone.

Question (4 marks) Discuss the advantages and disadvantages of moving from local storage to cloud storage.

Mark Scheme:

- **Advantage:** Scalability; users can pay for more space instantly without buying new hard drives.
 - **Advantage:** Accessibility; files can be reached from any location with internet.
 - **Disadvantage:** Security/Privacy; data is held by a third party and could be vulnerable to breaches.
 - **Disadvantage:** Connectivity; files cannot be accessed if the internet connection fails.
-

ON-PREMISES

Definition Data kept on company-owned hardware in their own buildings.

Explanation On-premises storage gives a company total physical control over its data, which is often required for high-security or top-secret information.

Scenario A local police station keeps its criminal evidence files on a server located in a locked room inside the station rather than using the cloud.

Question (3 marks) Explain one reason why a bank might prefer on-premises storage over cloud storage.

Mark Scheme:

- It provides total physical control over the hardware, reducing the risk of unauthorized third-party access.
 - It avoids ongoing subscription fees associated with cloud providers.
 - It ensures high-speed access to data without being limited by external internet bandwidth.
-

FILE ENCRYPTION

Definition Using an algorithm to scramble data into an unreadable form.

Explanation Encryption transforms "plaintext" (readable data) into "ciphertext" (unreadable code) so that even if the file is stolen, it cannot be understood without a key.

Scenario A journalist encrypts their interview notes so that if their laptop is stolen at the airport, the thief cannot read the sensitive information.

Question (3 marks) Describe the role of the "key" in the file encryption process.

Mark Scheme:

- The key is used by the algorithm to transform plaintext into ciphertext during encryption.
 - The same (or a related) key is required to reverse the process during decryption.
 - Without the correct key, the data remains scrambled and unreadable to unauthorized users.
-

ALGORITHM

Definition A finite set of predefined steps to solve a problem.

Explanation In security, an encryption algorithm is the mathematical formula (like AES or RSA) used to shuffle the data.

Scenario A programmer writes a new algorithm that follows ten specific mathematical steps to hide a user's password in the database.

Question (3 marks) Explain the function of an algorithm in symmetric encryption.

Mark Scheme:

- It provides the predefined steps to combine the plaintext with the secret key.
 - It generates the unreadable ciphertext based on those steps.
 - It ensures that the process can be perfectly reversed if the identical key is provided.
-

SYMMETRIC ENCRYPTION

Definition Uses the same key to both encrypt and decrypt data.

Explanation Symmetric encryption is very fast and efficient for large files, but it is risky because the sender and receiver must find a secure way to share the single key.

Scenario A company uses one master password (the key) to encrypt all its nightly database backups stored on an internal hard drive.

Question (3 marks) State one benefit and one major limitation of symmetric encryption.

Mark Scheme:

- **Benefit:** It is significantly faster and more efficient than asymmetric encryption for large datasets.
 - **Limitation:** Key exchange risk; if the single key is intercepted while being shared, all data is compromised.
 - **Benefit:** Highly secure against opportunistic "brute-force" attacks if the key is long and complex.
-

ASYMMETRIC ENCRYPTION

Definition Also known as public-key encryption; uses two related keys (Public and Private) that are mathematically linked but different.

Explanation The Public key can be given to anyone to encrypt a message, but only the specific owner's Private key can decrypt it, making it ideal for the internet.

Scenario When you send a message on WhatsApp, your phone uses the recipient's Public key to lock the message, and only their phone's Private key can open it.

Question (4 marks) Explain the process of sending a secure message using asymmetric encryption.

Mark Scheme:

- The sender obtains the recipient's Public Key, which can be shared openly.
 - The sender uses the Public Key + an algorithm to create the ciphertext.
 - The ciphertext is transmitted over the network (e.g., the internet).
 - The recipient uses their unique, secret Private Key to decrypt the message back into plaintext.
-

DIGITAL CERTIFICATES

Definition Document identifying the owner of a public key.

Explanation A digital certificate acts like an "ID card" for a website or user, proving that the Public key they are using actually belongs to them.

Scenario Your web browser checks a digital certificate when you visit www.paypal.com to make sure a hacker isn't pretending to be the bank.

Question (3 marks) Identify three components typically found within a digital certificate.

Mark Scheme:

- The owner's Public Key.
 - The name of the trusted Certificate Authority (CA) that issued it.
 - The validity period (expiry date) and the issuer's digital signature.
-

CERTIFICATE AUTHORITY (CA)

Definition A trusted third party that issues certificates as proof of identity.

Explanation CAs (like DigiCert or GoDaddy) verify that a business is legitimate before giving them a certificate, ensuring trust in online transactions.

Scenario A new online shop applies to a Certificate Authority to prove they are a real business so they can get the "Padlock" icon in the browser.

Question (3 marks) Explain the role of a Certificate Authority in establishing a secure HTTPS connection.

Mark Scheme:

- It acts as a trusted middleman to authenticate the identity of the website owner.
 - It digitally signs the certificate to verify it hasn't been tampered with.
 - Browsers use the CA's signature to confirm the website's public key is genuine during the key exchange.
-

SSL/TLS

Definition Secure version of the protocol used to transfer webpages / Networking protocol providing secure connections.

Explanation SSL/TLS uses a combination of asymmetric encryption (to share a key) and symmetric encryption (to send data) to keep web traffic private.

Scenario When you see <https://> in your browser, SSL/TLS is actively encrypting the credit card details you are typing so hackers cannot "sniff" the data.

Question (4 marks) Describe the "Handshake" process used by SSL/TLS to secure a connection.

Mark Scheme:

- **Exchange:** The browser requests the server's identification and receives its SSL certificate.
- **Verification:** The browser checks the certificate with a trusted Certificate Authority (CA).
- **Key Exchange:** The browser creates a temporary "session key" and encrypts it with the server's public key.
- **Transmission:** Both sides now use the faster symmetric encryption to send data using that session key.

PASSWORDS

Definition Knowledge-based authentication.

Explanation Passwords are the most common security layer, but their effectiveness depends entirely on their length and complexity to resist guessing.

Scenario A company policy requires all passwords to be at least 12 characters and include a mix of symbols to prevent brute-force attacks.

Question (3 marks) Explain how password complexity and length improve system security.

Mark Scheme:

- Increased length exponentially increases the time needed for a "brute-force" attack to guess every combination.
- Complexity (using symbols/cases) prevents simple "dictionary attacks" from working.
- It provides a baseline "knowledge-based" proof that the user is who they claim to be.

ACCESS CONTROL LISTS (ACLs)

Definition Lists of users/roles associated with specific database objects.

Explanation ACLs define exactly who is allowed to "Read," "Write," or "Delete" specific files or tables within a system.

Scenario An ACL for the "Salaries" folder allows the HR Manager to edit files but only allows regular employees to view their own PDF.

Question (3 marks) Describe how an administrator uses ACLs to manage database security.

Mark Scheme:

- They associate specific users or roles with specific database objects (like tables or views).
- They grant or deny specific "privileges" (e.g., SELECT, UPDATE, DELETE) for each user.

- This follows the principle of "least privilege," ensuring users only access what they need for their job.
-

ROW LEVEL SECURITY (RLS)

Definition Controlling access to individual rows based on conditions.

Explanation RLS allows different users to look at the *same* table but only see the specific rows that belong to them (e.g., a salesman seeing only his own customers).

Scenario In a multi-national hospital database, a doctor in Paris can only see rows where the `hospital_city` column is set to "Paris".

Question (3 marks) Explain one advantage of Row Level Security over standard table-level access.

Mark Scheme:

- It allows for much finer control, as access is based on the *content* of the data rather than the whole table.
 - It simplifies application development, as the database automatically filters the data for each user.
 - It prevents users from seeing sensitive records belonging to other departments or clients within the same table.
-

HOW TO ANSWER GUIDE

1. DEFINE (1-2 marks)

- **What to include:** Aim for the "textbook" definition. Always include the core mechanism (how it works) or the primary purpose (why we use it).
- **What to avoid:** Never use the word you are defining in the answer. Do not just give an example; give the rule.

- **Pro-tip:** For Topic 13, mention "hardware abstraction" for virtualisation or "independent nodes" for distributed systems.

2. EXPLAIN (3-4 marks)

- **Structure:** Use **Point -> Evidence -> Impact**.
- **Keywords:** Use "Therefore," "Consequently," or "This leads to..." to link your points.
- **Focus:** The examiner wants to see you understand the *consequence*. For example, don't just say virtualisation is fast; say "It improves scalability because servers can be created via software in seconds, leading to reduced downtime".

3. DISCUSS/EVALUATE (6-8 marks)

- **Structure:** **Introduction -> Advantages -> Disadvantages -> Conclusion**.
- **Scenario Use:** You *must* reference the scenario. If the prompt is about a "Hospital," mention "Patient data privacy" and "Doctor's access," not just "users".
- **The "Balanced View":** For Topic 13, often compare VMs vs Containers or Cloud vs On-premises. Weigh the cost against the security or performance benefits.

4. COMMON MISTAKES (TOPIC 13)

- **VMs vs Containers:** Many students think containers have their own OS. They don't! Containers share the host kernel.
- **Symmetric vs Asymmetric:** Students often forget that Symmetric is *faster* but less secure for transmission due to the "Key Exchange" problem.
- **HCI vs UI:** Don't use them interchangeably. UI is the *thing* (buttons); HCI is the *study* of the interaction.
- **Distributed Systems:** Don't confuse them with standard networks. A distributed system *must* have nodes working together for a "common goal".

TOPIC 14

IT Systems in Organisations

Definition IT systems are indispensable tools used to monitor production, improve productivity, and provide workers with access to business-critical information.

Explanation These systems ensure day-to-day operations are completed effectively for management, employees, and clients by supporting core functions.

Scenario A textile factory implements a centralized IT system to track yarn usage across the production floor while giving managers real-time access to stock levels.

Question (3-4 marks) Explain why a textile factory would consider its IT inventory system to be "indispensable" to its operations.

Mark Scheme:

- It allows management to monitor production levels and material usage in real-time.
 - It improves overall productivity by ensuring workers have the data needed to perform tasks without delay.
 - It provides access to business-critical information such as stock thresholds to prevent production stops.
-

Business-Critical

Definition Data, information, or tasks/operations essential for an organisation to continue to operate.

Explanation If a business-critical system fails, the organization may face immediate operational stoppage or significant financial loss.

Scenario For an online bank, the server processing customer fund transfers is a business-critical component because its failure stops all primary services.

Question (3-4 marks) Identify and explain one reason why a payment processing system is classified as business-critical for a retail company.

Mark Scheme:

- A payment system is essential for the organisation to continue to operate as it facilitates revenue collection.
 - Failure of this task would stop day-to-day operations and prevent customers from completing purchases.
 - It involves information that is indispensable for maintaining financial records and legal compliance.
-

Automation

Definition Replacing repetitive, simplistic, or laborious tasks to free employees for strategic work.

Explanation Automation increases efficiency by handling mundane tasks like data entry or order generation through software rather than human effort.

Scenario A warehouse uses software to automatically generate purchase orders whenever sensors detect that shelf stock has fallen below a certain weight.

Question (3-4 marks) Explain one benefit to employees when an organisation introduces automation for data entry tasks.

Mark Scheme:

- Automation replaces repetitive and laborious manual entry tasks.
 - This frees up the employees to focus on more strategic or complex work for the business.
 - It reduces the likelihood of human error in simple tasks, improving the quality of the data the employees work with.
-

Monitoring and Maintenance

Definition Continuous performance tracking to spot potential issues before they cause downtime or data loss.

Explanation Proactive tracking allows IT teams to identify hardware failures or software bugs before they impact the end-user.

Scenario An ISP uses monitoring software to track server temperatures and network traffic loads to prevent hardware melting or service outages.

Question (3-4 marks) Describe the role of continuous performance tracking in an organisation's IT strategy.

Mark Scheme:

- It allows the organization to monitor system health and performance levels constantly.
 - This helps IT staff spot potential technical issues early.
 - Corrective action can be taken before the issues cause significant downtime or loss of data.
-

Customer Service

Definition Using self-service portals, chatbots, and knowledge bases to provide 24/7 support.

Explanation IT-driven customer service improves satisfaction by providing instant responses to common queries without requiring human staff.

Scenario A global airline uses an AI chatbot on its website to help customers rebook flights and check luggage allowances at any time of night.

Question (3-4 marks) Explain how the use of a "knowledge base" can improve the quality of customer service in an organisation.

Mark Scheme:

- A knowledge base provides customers with 24/7 access to information and support.
 - It allows for self-service, enabling customers to find answers to common queries quickly without waiting for a human agent.
 - This improves the consistency of the information provided to all stakeholders.
-

ERP (Enterprise Resource Planning)

Definition Systems that integrate finance, HR, and supply chain into a single centralised system.

Explanation ERP systems eliminate data silos by ensuring all departments use the same data source for resource management.

Scenario A construction firm uses ERP software so that when a new project is signed, the HR, finance, and logistics departments all see the same updated resource requirements.

Question (3-4 marks) Explain why a large multinational corporation would choose to implement an ERP system.

Mark Scheme:

- An ERP integrates different business functions like finance and HR into a single centralised system.
- It allows for better resource management by providing a unified view of the entire supply chain.
- It improves efficiency by ensuring data is shared seamlessly across different departments.

Collaboration

Definition Cloud platforms that enable distributed teams to work simultaneously.

Explanation Collaboration tools allow multiple users to edit documents and share ideas in real-time regardless of their physical location.

Scenario A design team spread across three different time zones uses a cloud-based document editor to finalize a marketing brochure together.

Question (3-4 marks) Describe how cloud platforms facilitate collaboration among distributed teams.

Mark Scheme:

- Cloud platforms allow team members to work on the same project or document simultaneously.
- They enable distributed teams to stay synchronized regardless of their physical location.
- They often include tools for instant communication to resolve project issues in real-time.

Version Control

Definition Tracking and managing changes in documents/software to prevent data loss or conflicting edits.

Explanation It ensures that if multiple people edit a file, a history of changes is kept so previous versions can be restored if an error occurs.

Scenario A software developer uses version control to ensure that their new code doesn't overwrite a colleague's work on the same application module.

Question (3-4 marks) Explain one reason why version control is essential when multiple employees are working on the same software project.

Mark Scheme:

- Version control tracks and manages every individual change made to the software files.
- It prevents conflicting edits where one employee's work might accidentally overwrite another's.
- It allows the team to revert to a previous working version if the new edits introduce errors or data loss.

Workflow

Definition A sequence of tasks/processes carried out by groups/individuals to complete an objective.

Explanation IT systems streamline workflows by automating the transition of a project from one stage to the next.

Scenario An insurance company's workflow system automatically sends a claims form to the legal department once the initial adjuster has approved the damage.

Question (3-4 marks) Explain how an automated workflow system can improve the efficiency of service delivery.

Mark Scheme:

- It defines a clear sequence of tasks that must be followed to complete an objective.
 - It streamlines the process by moving tasks between individuals or departments automatically.
 - This reduces delays and ensures consistent processing of every service request.
-

Transaction Processing

Definition The management of individual tasks involving data changes in a system, usually financial or produce-related.

Explanation Transaction processing focuses on maintaining data integrity and accuracy during high-volume operations.

Scenario A supermarket's database updates the exact millisecond a customer buys a loaf of bread to ensure the stock count is perfectly accurate.

Question (3-4 marks) Describe the primary goal of a transaction processing system in a retail environment.

Mark Scheme:

- The primary goal is to maintain the integrity and accuracy of the data within the system.
 - It ensures that individual tasks involving data changes, such as a sale, are recorded correctly.
 - It makes data available for other processes like real-time inventory management.
-

Back-office Processes

Definition Administrative and logistical tasks supporting a business without direct customer interaction.

Explanation These processes are "behind the scenes" but are vital for the operational success of the company's front-end services.

Scenario A shipping company's system for calculating fuel costs for the next month is a back-office process that customers never see.

Question (3-4 marks) Distinguish between back-office processes and customer-facing IT systems.

Mark Scheme:

- Back-office processes involve administrative or logistical tasks, such as payroll or inventory planning.
 - They do not require direct interaction with the customer.
 - Customer-facing systems, like EPOS or self-service portals, are designed for direct stakeholder engagement.
-

EPOS (Electronic Point of Sale)

Definition Specific application used for sales processing, inventory management, and gathering customer insights.

Explanation EPOS systems integrate the physical checkout process with digital databases to automate stock reordering and loyalty tracking.

Scenario When a barista scans a coffee at the till, the EPOS system subtracts one bag of beans from the digital inventory and adds points to the customer's app.

Question (3-4 marks) Explain two ways an EPOS system assists a manager in running a retail store.

Mark Scheme:

- It provides real-time inventory management by tracking stock levels and automatically reordering items.
 - It gathers customer insights by identifying popular items and peak sales times to help with staffing decisions.
 - It processes payments efficiently, supporting card, contactless, and smartphone transactions.
-

Atomicity

Definition Transactions cannot be split; treated as one indivisible unit (complete fully or not at all).

Explanation This prevents partial data updates which would lead to inconsistencies, such as money leaving one account but not arriving in another.

Scenario In a bank transfer, atomicity ensures that if the system crashes after debiting Account A but before crediting Account B, the whole transaction is rolled back.

Question (3-4 marks) Explain why atomicity is a critical principle in a financial transaction processing system.

Mark Scheme:

- Atomicity ensures that a transaction is treated as a single, indivisible unit.
 - It guarantees that the transaction either completes in its entirety or does not happen at all.
 - This prevents errors where only part of a financial update is recorded, maintaining data integrity.
-

Consistency

Definition System moves from one valid state to another (e.g., stock levels adjust correctly based on the order).

Explanation Consistency ensures that all predefined rules—such as data types and constraints—are followed after every transaction.

Scenario If a customer buys 5 shirts, the consistency rule ensures the inventory count drops by exactly 5, not a random or invalid number.

Question (3-4 marks) Describe how the "consistency" principle helps maintain the reliability of a database.

Mark Scheme:

- Consistency ensures the system only moves from one valid state to another valid state.

- It ensures that all data follows set rules, such as updating stock levels correctly based on an order.
 - This prevents the database from containing conflicting or illogical information.
-

Isolation

Definition Simultaneous transactions do not interfere with each other.

Explanation Isolation prevents "race conditions" where two users trying to access the same resource at once cause a data error.

Scenario Isolation ensures that if two people click "book" on the last cinema seat at the exact same time, only one transaction succeeds and the other is told the seat is gone.

Question (3-4 marks) Explain how "isolation" manages multiple users accessing a booking system at the same time.

Mark Scheme:

- Isolation ensures that simultaneous transactions are processed as if they occurred sequentially.
 - It prevents transactions from interfering with one another during processing.
 - This ensures that a single resource, like a seat, cannot be double-booked by two different users.
-

Durability

Definition Confirmed details/changes are permanent and stored securely, even after unexpected events.

Explanation Once a transaction is committed to the database, it must remain there even in the event of a power cut or system crash.

Scenario After a customer receives a "payment successful" message, durability ensures their order is saved in the database even if the store's server loses power a second later.

Question (3-4 marks) Explain the importance of "durability" in a transaction processing system.

Mark Scheme:

- Durability ensures that once a transaction is confirmed, the changes are made permanent.
 - The data is stored securely so it cannot be lost.
 - This ensures data remains intact even if the system suffers an unexpected event like a crash or power failure.
-

Bacs (Bacs Payment Schemes Limited)

Definition UK organisation managing direct debit and direct credit; requires customer authorisation for bank account deductions.

Explanation Bacs provides a standard, secure framework for processing high-volume automated payments between bank accounts in the UK.

Scenario A gym uses Bacs to automatically collect monthly membership fees from its 500 members' bank accounts on the first of every month.

Question (3-4 marks) Explain the requirements for an organisation to use the Bacs system to collect payments.

Mark Scheme:

- The organisation must be operating within the UK as Bacs is a UK-specific scheme.
 - They must obtain explicit customer authorisation before any deductions are made.
 - They use the system to manage automated tasks like direct debits or direct credits.
-

CRM (Customer Relationship Management)

Definition Software tools designed to help businesses manage interactions/relationships with customers and stakeholders.

Explanation CRM systems centralize customer data to help businesses provide personalized experiences and improve retention.

Scenario A car dealership uses its CRM to automatically send a "happy birthday" email with a 10% discount on the customer's next oil change.

Question (3-4 marks) Explain how a CRM system can help a business increase its engagement with customers.

Mark Scheme:

- It allows the business to personalise interactions by using purchase history and preferences.
 - It can synchronise marketing events, sending targeted ads or vouchers through various media.
 - By tracking enquiries and issues, it ensures a satisfactory follow-up, building a positive relationship.
-

Loyalty Schemes

Definition Tracking purchases to award points/rewards for future discounts.

Explanation Loyalty schemes use CRM data to incentivize customers to return by offering tangible rewards for their repeat business.

Scenario A coffee shop's digital app tracks every 10th purchase and automatically issues a voucher for a free latte to the customer's account.

Question (3-4 marks) Describe the role of "loyalty schemes" within a Customer Relationship Management (CRM) strategy.

Mark Scheme:

- Loyalty schemes track customer purchases to identify buying patterns.
 - They are used to award points or rewards that provide discounts on future sales.
 - The goal is to secure return custom and improve customer retention rates.
-

Upselling

Definition Identifying patterns to recommend "add-ons" to orders.

Explanation Upselling uses CRM analytics to suggest complementary products that increase the total value of a customer's transaction.

Scenario While a customer is buying a camera online, the CRM system prompts them to add a protective carrying case and a high-speed SD card to their basket.

Question (3-4 marks) Using an example, explain how a CRM system can facilitate "upselling."

Mark Scheme:

- The system identifies buying trends and patterns from existing customer data.
 - It then automatically recommends "add-ons" that complement the current purchase.
 - For example, recommending a specific headset to a customer who is buying a laptop.
-

MIS (Management Information Systems)

Definition Systems/processes providing managers with information to run an organisation and make data-driven decisions.

Explanation MIS aggregates data from various sources (like EPOS and CRM) and presents it through dashboards to simplify complex decision-making.

Scenario A retail manager uses an MIS dashboard to compare this week's sales figures against the same week last year to decide on staffing levels.

Question (3-4 marks) Explain how an MIS can help a manager reduce complexity when making business decisions.

Mark Scheme:

- MIS gathers data from multiple internal and external sources, such as EPOS and market research.
 - It applies processing like cleansing and aggregation to make the data useful.
 - It removes complexity from large datasets by presenting high-quality insights via query tools and dashboards.
-

ITS (Intelligent Transportation Systems)

Definition ITS aim to improve the planning and operation of public and private transportation using sensors, cameras, and monitoring systems.

Explanation ITS uses real-time data to optimize routes and schedules, reducing fuel consumption and travel times.

Scenario A city uses ITS to adjust the timing of traffic lights automatically when cameras detect a large buildup of cars on a main road.

Question (3-4 marks) Describe how an ITS can improve the fuel efficiency of a transportation network.

Mark Scheme:

- It uses sensors and real-time traffic data to calculate optimal routes.
 - This reduces the time vehicles spend idling in traffic or congestion.
 - By minimizing travel time, the system directly reduces overall fuel consumption.
-

Telematics

Definition An in-vehicle system using GPS and vehicle diagnostics to provide detailed, real-time information.

Explanation Telematics allows fleet managers to track locations and monitor driver behavior, such as speeding or harsh braking.

Scenario A delivery company uses telematics to see that one of its vans has been idling for 20 minutes and sends the driver an alternative route to avoid a road closure.

Question (3-4 marks) Explain how a manager can use "telematics" to improve driver safety within a delivery fleet.

Mark Scheme:

- Telematics systems use GPS and diagnostics to provide real-time information on the vehicle.

- They allow managers to monitor driver behaviour, such as sudden braking or speeding.
 - Managers can use this data to provide proactive training or feedback to improve safe driving habits.
-

Dynamic Scheduling

Definition Adjusting bus/train times based on passenger numbers and congestion.

Explanation Unlike fixed timetables, dynamic scheduling uses live data to provide transport services exactly when and where they are needed.

Scenario A bus company sees a sudden surge of 50 passengers at a specific stop via an app and immediately dispatches an extra bus to that location.

Question (3-4 marks) Contrast dynamic scheduling with traditional fixed timetabling in public transport.

Mark Scheme:

- Fixed timetabling uses historical data to set frequency, while dynamic scheduling uses real-time data.
 - Dynamic scheduling allows for quick adjustments based on current passenger demand or congestion.
 - This results in a more flexible service that can respond immediately to incidents or surges.
-

Expert Systems

Definition An IT system that supports decision-making in a specific area of knowledge.

Explanation Expert systems mimic human reasoning by applying programmed rules to a specific set of facts to provide solutions.

Scenario A hospital uses an expert system where a nurse inputs a patient's symptoms, and the software suggests three possible diagnoses based on medical rules.

Question (3-4 marks) Explain how an expert system differs from a typical machine learning AI.

Mark Scheme:

- Expert systems rely on pre-programmed rules and facts encoded by human experts.
 - They have a narrow scope and only work within very strict parameters.
 - Unlike AI, they are static and their knowledge base does not automatically improve through new data.
-

Knowledge Base

Definition A large repository of vast information/expertise related to a specific domain.

Explanation The knowledge base acts as the "memory" of an expert system, containing the facts and rules that the system uses to solve problems.

Scenario In a technical support expert system, the knowledge base contains all known error codes and the specific repair steps for every laptop model the company makes.

Question (3-4 marks) Describe the role of the "knowledge base" within an expert system.

Mark Scheme:

- It serves as a large repository of information and expertise for a specific domain.
 - It contains the programmed rules and facts that represent human expert knowledge.
 - It provides the raw information that the system processes to draw conclusions.
-

Inference Engine

Definition The decision-making part that processes knowledge base info to draw conclusions and recommend solutions.

Explanation The inference engine acts as the "brain," applying logical reasoning to the facts in the knowledge base to answer a user's query.

Scenario When a mechanic enters "engine smoke" into a diagnostic tool, the inference engine searches the knowledge base's rules to conclude that the oil gasket is likely leaking.

Question (3-4 marks) Explain the relationship between the knowledge base and the inference engine in an expert system.

Mark Scheme:

- The knowledge base contains the repository of facts and expert rules.
 - The inference engine acts as the decision-making component that processes this info.
 - It applies the rules from the knowledge base to the user's specific data to recommend a solution.
-

IT Governance

Definition The pre-defined processes, structures, and mechanisms ensuring IT resources are used appropriately and effectively.

Explanation Governance provides a framework to ensure that IT investments deliver value and that risks are managed according to business goals.

Scenario A multinational bank establishes an IT governance board to approve all new software purchases to ensure they meet security and legal standards.

Question (3-4 marks) Explain why IT governance is necessary for a large organisation.

Mark Scheme:

- It ensures IT activities are strategically aligned with the goals of the business.
 - It provides a structure for identifying and mitigating potential risks to IT systems.
 - It ensures compliance with legal regulations, such as data protection and privacy laws.
-

Strategic Alignment

Definition IT activities must align with business strategies and goals.

Explanation This principle ensures that the IT department isn't just buying technology for its own sake, but is supporting the company's long-term objectives.

Scenario If a company's goal is to become the fastest delivery service, its IT strategic alignment involves investing in real-time route optimization rather than fancy new office laptops.

Question (3-4 marks) Explain the importance of "strategic alignment" in IT governance.

Mark Scheme:

- It ensures that IT resources are used to support the overarching goals of the business.
 - It prevents wasteful spending on technology that does not contribute to business success.
 - It helps managers prioritize IT projects that offer the most value to the organization.
-

Risk Management

Definition Identifying potential risks to IT systems and mitigation methods.

Explanation Risk management involves assessing the probability and impact of threats to ensure the business can survive technical failures.

Scenario An IT manager identifies that their single internet connection is a risk and mitigates it by installing a second backup line from a different provider.

Question (3-4 marks) Describe the steps an organisation should take to manage IT risks effectively.

Mark Scheme:

- The organisation must first identify potential threats, such as cyber-attacks or human error.
 - They must then perform an impact assessment to evaluate the severity of those risks.
 - Finally, they must develop and implement mitigation strategies to reduce the likelihood or harm of the risk.
-

Risk Score

Definition $\text{RiskScore} = \text{Probability} \times \text{Severity}$.

Explanation Calculated using scales (typically 1-5), the risk score helps organizations prioritize which threats need the most urgent attention.

Scenario A risk with a probability of 5 (almost certain) and a severity of 4 (major) gets a score of 20, marking it as a "severe" priority.

Question (3-4 marks) Explain how an organisation uses a "Risk Score" to prioritize its IT security budget.

Mark Scheme:

- The score is calculated by multiplying the probability of a risk by its severity.
 - This provides a numerical value that represents the level of risk (e.g., severe vs. adequate).
 - The organisation can then allocate more budget to mitigating risks with higher scores.
-

Business Continuity

Definition Ensuring core functions continue during a data breach or system failure through fallback systems and specific procedures.

Explanation Business continuity is about staying operational *during* a crisis, rather than just fixing things *after* it.

Scenario A hospital has a business continuity plan that involves switching to paper records immediately if the digital patient database goes offline.

Question (3-4 marks) Explain the difference between business continuity and disaster recovery.

Mark Scheme:

- Business continuity focuses on ensuring core functions can continue *during* an ongoing failure or breach.
 - It uses fallback systems or manual procedures to maintain operations.
 - Disaster recovery specifically focuses on restoring data and systems *after* a serious issue has occurred.
-

Disaster Recovery

Definition Policies for restoring data as quickly and completely as possible after a serious issue (natural disaster, cyber attack).

Explanation Disaster recovery focuses on the technical restoration of IT systems back to their normal state after an outage.

Scenario After a flood destroys the main server room, the company triggers its disaster recovery plan to restore all data from an off-site cloud backup.

Question (3-4 marks) Identify three key requirements that should be included in a disaster recovery plan.

Mark Scheme:

- An inventory of all critical data and systems that need protection.
- Clearly defined recovery objectives, such as RTOs and RPOs.
- Detailed backup and recovery procedures, including instructions on how to restore the data.

Acceptable Use Policy (AUP)

Definition Defines acceptable behaviours for using IT resources (internet, email, sensitive info) and consequences for failure.

Explanation An AUP protects the organization legally and ensures that employees do not misuse company technology for personal or harmful reasons.

Scenario An employee's contract includes an AUP stating they cannot use company email to sign up for personal social media accounts or download unlicensed software.

Question (3-4 marks) Explain one benefit to an employer of having an Acceptable Use Policy (AUP) in place.

Mark Scheme:

- It defines clear expectations for how employees should behave when using IT resources.
- it provides a legal framework for the company to take action if an employee misuses sensitive information.

- It helps prevent security risks, such as employees visiting harmful websites or downloading malware.
-

Phased Changeover

Definition Implementing the system gradually in stages or modules (e.g., Finance first, then Inventory).

Explanation Phased changeover is lower risk because if a problem occurs, it only affects the specific module currently being implemented.

Scenario A school decides to move to a new IT system by implementing the "Attendance" module in September and the "Grades" module in January.

Question (3-4 marks) State one advantage and one disadvantage of using a phased changeover methodology.

Mark Scheme:

- Advantage: Reduces risk as any issues are restricted to only one area of the business.
 - Disadvantage: It takes much longer to realize the full benefits of the complete system.
 - Disadvantage: It requires the new modules to be compatible with the old system during the transition.
-

Direct Changeover

Definition Replacing the old system with the new one at a specific cut-off point.

Explanation Often called the "Big Bang" approach, this method is fast but highly risky as there is no fallback if the new system fails.

Scenario At midnight on Sunday, a small cafe turns off its old cash register and starts using its new tablet-based EPOS system exclusively.

Question (3-4 marks) Explain why a large, business-critical organisation would likely avoid a direct changeover.

Mark Scheme:

- Direct changeover is high risk because there is no fallback system if the new one fails.
 - Any system error could lead to a complete stoppage of business-critical operations.
 - There is a significant productivity drop as all employees must face the learning curve at the same time.
-

Parallel Changeover

Definition Running both systems simultaneously for a set period.

Explanation This is the safest changeover method because the old system remains fully operational as a fallback while the new one is tested.

Scenario A legal firm processes all its new cases on the new cloud system while also entering them into the old desktop database for the first month to ensure the numbers match.

Question (3-4 marks) Explain why an organisation might find a parallel changeover to be expensive.

Mark Scheme:

- It requires running two systems simultaneously, which increases hardware and software costs.
 - It leads to an increased and duplicated workload for employees who must enter data twice.
 - It requires more IT staff resources to maintain and compare both systems during the set period.
-

Pilot Changeover

Definition Full system implemented within a limited area or group first.

Explanation A pilot changeover allows for real-world testing in a controlled environment before rolling the system out to the whole company.

Scenario A retail chain tests its new inventory system only in its London branch before deciding whether to install it in all 50 stores nationwide.

Question (3-4 marks) Explain one benefit of using a pilot changeover for a large retail chain.

Mark Scheme:

- It allows for testing of the full system in a real-world setting.
 - Any risks or bugs identified are restricted to the limited pilot area, preventing a company-wide failure.
 - Feedback from the pilot group can be used to improve the system before the final roll-out.
-

Perfective Maintenance

Definition Identifying and implementing improvements to working systems to increase efficiency or functionality.

Explanation Perfective maintenance is proactive and focuses on making a "good" system even better or faster for the users.

Scenario After feedback that the checkout process is too slow, a developer rewrites the code to reduce the number of clicks required to finish a sale.

Question (3-4 marks) Distinguish between perfective and corrective maintenance.

Mark Scheme:

- Perfective maintenance is proactive and aims to increase the efficiency or functionality of a working system.
 - Corrective maintenance is reactive and focuses on fixing bugs, errors, or defects that have been identified.
 - Perfective maintenance adds value, while corrective maintenance restores expected performance.
-

Adaptive Maintenance

Definition Modifying systems to accommodate changes in the external environment.

Explanation This is necessary when factors outside the company's control, like new laws or hardware updates, change.

Scenario A payroll company updates its software code to reflect the new tax percentages recently announced by the government.

Question (3-4 marks) Explain one scenario where an organisation would be forced to carry out adaptive maintenance.

Mark Scheme:

- Adaptive maintenance is required when there is a change in the external environment.
 - For example, if the government introduces new legal regulations or tax laws.
 - It can also occur when the business moves to new hardware or a new operating system.
-

Corrective Maintenance

Definition Reactive process of fixing errors, bugs, or defects identified through user feedback or abnormal behaviour.

Explanation Corrective maintenance is a necessity for system stability and must always be followed by rigorous testing to ensure the fix works.

Scenario An employee reports that the "Print" button in the accounting software crashes the computer, so the IT team writes a patch to fix the bug.

Question (3-4 marks) Explain why corrective maintenance must be followed by "regression testing."

Mark Scheme:

- Corrective maintenance is the process of fixing identified bugs or errors.
 - Regression testing is a mandatory rule that must be followed after a fix is applied.
 - It ensures that the new changes have not introduced any new issues or broken existing functionality.
-

Regression Testing

Definition Testing to ensure no new issues were introduced after system changes.

Explanation It is a quality assurance step performed after any form of maintenance to verify that the system's "old" features still work correctly.

Scenario After updating the login screen to be faster, the tester performs regression testing to make sure users can still actually log in to their accounts.

Question (3-4 marks) Explain the role of regression testing in the system maintenance cycle.

Mark Scheme:

- It is performed after corrective, adaptive, or perfective maintenance has been carried out.
 - Its role is to verify that the changes made to the system have not caused unexpected errors elsewhere.
 - It ensures that the overall integrity and stability of the system are maintained.
-

Archiving

Definition Storing historical data no longer in continuous use securely while maintaining accessibility.

Explanation Archiving frees up space on primary production servers while ensuring data remains available for legal or analytical reasons.

Scenario A law firm moves its digital files for cases that were closed more than five years ago from its main server to a cheaper, long-term storage vault.

Question (3-4 marks) Explain two reasons why an organisation would choose to archive data rather than delete it.

Mark Scheme:

- Compliance: Legal or regulatory requirements may mandate that records are retained for specific periods.
- Historical Analysis: The data remains valuable for identifying long-term trends or planning.

- Auditors: It provides evidence of data integrity and business compliance if an investigation occurs.
-

RTO (Recovery Time Objective)

Definition The timeframe in which data must be recovered/restored.

Explanation RTO defines the "maximum allowable downtime" a business can tolerate before serious damage occurs.

Scenario An online retailer sets an RTO of 2 hours, meaning their technical team must have the website back up and running within 2 hours of any crash.

Question (3-4 marks) Explain how an RTO of "zero" would affect an organisation's IT infrastructure choices.

Mark Scheme:

- An RTO of zero means data and systems must be recovered/restored immediately with no downtime.
 - This would require expensive high-availability systems, such as mirrored servers in different locations.
 - It would also require constant real-time monitoring and automated failover procedures.
-

RPO (Recovery Point Objective)

Definition The maximum amount of data (in time) an organisation is willing to lose.

Explanation RPO determines how frequently backups must be made; an RPO of 1 hour means backups must occur at least every hour.

Scenario A bank with an RPO of 15 minutes performs a full database backup four times every hour to ensure they never lose more than 15 minutes of customer transactions.

Question (3-4 marks) Explain the relationship between an RPO and the frequency of data backups.

Mark Scheme:

- RPO defines the maximum interval of data loss an organisation is willing to tolerate.
 - The frequency of backups must be at least as short as the RPO.
 - For example, an RPO of one hour mandates that data is backed up at least once every hour to ensure compliance.
-

HOW TO ANSWER GUIDE

1. DEFINE (1-2 marks)

- **What to include:** Use the exact terminology from the textbook. For example, if defining "Atomicity," use words like "indivisible" and "not split".
- **What to avoid:** Do not use the word itself in the definition. Avoid vague descriptions; be technically precise.

2. EXPLAIN (3-4 marks)

- **Structure:** Start with a definition, then provide a "chain of reasoning" (Why/How).
- **Keywords:** Use connective words like "Therefore," "Consequently," or "This leads to." For Topic 14, always link back to "efficiency," "data integrity," or "business-critical" needs.

3. DISCUSS/EVALUATE (6-8 marks)

- **Structure:** Introduction -> Points for (Pros) -> Points against (Cons) -> Conclusion/Judgment.
- **Scenario Use:** The examiner wants you to apply the terms to the specific business in the prompt. Don't just say "Parallel changeover is safe"; say "Parallel changeover is safe for this hospital because if the new patient database fails, doctors can still access the old one".
- **Judgment:** Always end with a recommendation based on the scenario's specific risks and budget.

4. Common mistakes on Topic 14

- **Mixing up ACID terms:** Students often confuse "Consistency" (valid states) with "Isolation" (preventing interference).

- **Vague Maintenance Definitions:** Ensure you can distinguish between "Adaptive" (external environment changes) and "Perfective" (improving internal performance).
- **RTO vs RPO:** Remember: **RTO** is about **Time** (downtime); **RPO** is about **Point** (data loss).
- **Expert Systems vs AI:** Expert systems are **static** and rule-based; they do not "learn" from new data like machine learning does.

TOPIC 15

Project Management

Definition The process of overseeing and delivering a project by managing human and physical resources to meet specific objectives within a set time, budget, and scope.

Explanation It functions as a framework to break complex IT tasks into manageable stages, ensuring the final product reaches quality standards without exceeding resources.

Scenario A software house is tasked with developing a new inventory system for a global retailer within six months.

Question (3-4 marks) Explain two reasons why a project manager is essential for the success of this inventory system project.

Mark Scheme: • To manage resource allocation, ensuring developers and hardware are used efficiently to meet the deadline. • To perform risk mitigation, identifying potential technical issues early and creating contingency plans. • To coordinate communication between stakeholders and the development team to ensure goals are met.

Project Manager

Definition The individual responsible for overseeing and delivering a project.

Explanation The project manager acts as the central point of accountability, balancing time, budget, and scope while managing the team.

Scenario Sarah has been appointed to lead a team of ten developers to create a mobile banking app.

Question (3-4 marks) Describe the role of Sarah as the project manager in ensuring the mobile banking app remains within its original scope.

Mark Scheme: • Sarah must implement change management procedures to evaluate and prioritise any new feature requests. • She must resist "feature creep" by preventing the continual addition of non-essential features that could delay the launch. • She must monitor progress against milestones to identify and resolve bottlenecks in development.

Stakeholder

Definition An individual or group with a particular interest in or link to a project.

Explanation Engaging stakeholders is critical because their requirements define the project's success and they provide essential feedback during development.

Scenario A local council is installing a new public Wi-Fi network in the town centre.

Question (3-4 marks) Identify one internal and one external stakeholder for this project and explain their interest.

Mark Scheme: • Internal: Council IT department/management who are responsible for maintaining the system's performance. • External: Local residents/businesses who will use the service and expect high reliability and accessibility. • External: Government agencies who may have interest in compliance or data protection regulations.

Mitigation

Definition An action or strategy intended to reduce the seriousness or harm caused by a potential risk.

Explanation Mitigation involves proactive planning to minimize the impact of identified risks, such as technical failures or staff shortages.

Scenario During a project to migrate data to a new server, the team identifies a risk that the old hardware might fail during the transfer.

Question (3-4 marks) Explain how the project team could implement a mitigation strategy for the identified hardware risk.

Mark Scheme: • The team could perform a full system backup before starting the migration to prevent permanent data loss. • They could schedule the migration during low-traffic hours to minimize the impact on users if a failure occurs. • They could have a secondary "fallback" server ready to take over immediately if the primary hardware fails.

Feature Creep

Definition The continual addition of new features to a product during development that negatively affects deadlines and budgets.

Explanation Feature creep is a major risk because it destabilizes project planning and often leads to missed "time to market" deadlines.

Scenario While building a simple calculator app, the client repeatedly asks to add a currency converter, a graphing tool, and social media sharing.

Question (3-4 marks) Describe the impact that feature creep would have on the development of this calculator app.

Mark Scheme: • It would likely cause the project to exceed its original budget as more developer hours are required. • It could lead to a missed deadline (time to market) as new features require additional design and testing phases. • It may lead to a loss of focus on the core functionality, potentially reducing the overall quality of the app.

Quality Assurance (QA)

Definition The process of ensuring the desired level of performance, design, and reliability is maintained throughout the project lifecycle.

Explanation QA is an ongoing process that involves setting standards and carrying out regular reviews to ensure the final product is fit for purpose.

Scenario A game development studio uses a dedicated team to play through every level of a new game to find bugs before release.

Question (3-4 marks) Explain why Quality Assurance should be maintained throughout the entire project lifecycle rather than just at the end.

Mark Scheme: • It allows for early detection of design flaws, which are cheaper and easier to fix than at the final stage. • It ensures that each phase (e.g., design, coding) meets specific quality standards before moving to the next. • It helps maintain stakeholder confidence by providing evidence of consistent progress and reliability.

Project Lifecycle

Definition The main phases of a project (initiation, planning, execution, monitoring, closure).

Explanation The lifecycle provides a structured path for project management, ensuring no critical steps like testing or evaluation are skipped.

Scenario A company is using a five-stage model to develop a new payroll system.

Question (3-4 marks) Identify three phases of the project lifecycle and describe a task associated with each.

Mark Scheme: • Planning: Outlining the project scope, resource requirements, and setting SMART targets. • Execution: The actual coding of the software or building of the IT infrastructure. • Monitoring: Tracking progress against milestones and monitoring the budget spend.

SMART Targets

Definition An approach to setting goals that ensures they can be used effectively to monitor and evaluate a project.

Explanation SMART targets turn vague ideas into concrete objectives that can be used to hold teams accountable and measure success.

Scenario A team wants to "make the website better" and decides to rewrite this as a SMART target.

Question (3-4 marks) Explain how using SMART targets helps a project manager evaluate the success of a project.

Mark Scheme: • It provides a "measurable" way to quantify success, such as a specific percentage increase in speed. • It includes "time-bound" deadlines, allowing the manager to see if milestones were met on schedule. • It ensures goals are "specific," making it clear to all stakeholders exactly what the project intends to achieve.

Specific (SMART)

Definition Goals must be clear and well-defined for easy tracking.

Explanation Specificity avoids ambiguity, ensuring every team member understands exactly what is required for a task to be complete.

Scenario Instead of saying "improve security," a manager specifies "implement 256-bit AES encryption for all stored customer data."

Question (3-4 marks) Explain why a "specific" target is more effective than a general one in IT project management.

Mark Scheme: • It removes ambiguity, ensuring developers know exactly which protocols or features to implement. • It allows for more accurate resource planning, as the team can identify specific tools or skills needed. • It makes final evaluation easier because the project manager can check if the specific requirement was met.

Measurable (SMART)

Definition There must be a way to quantify progress or success.

Explanation Measurement allows for objective assessment, removing guesswork from whether a target has been reached.

Scenario A database project sets a target to "reduce query response time by at least 40%."

Question (3-4 marks) Describe one way progress can be "measured" during the development of a new database system.

Mark Scheme: • By timing how long specific queries take to execute and comparing them against the original performance. • By tracking the number of bugs identified and resolved during the testing phase. • By calculating the percentage of the total project budget that has been spent at specific milestones.

Achievable (SMART)

Definition Goals must be attainable; large goals should be broken into manageable sub-tasks.

Explanation Setting achievable goals prevents team burnout and ensures the project does not fail due to unrealistic expectations.

Scenario A small startup decides not to build a full social media platform in one month and instead focuses on building a secure login module first.

Question (3-4 marks) Explain why it is important for project targets to be "achievable" for the development team.

Mark Scheme: • Unrealistic goals can lead to team burnout and low morale if they feel they cannot succeed. • It ensures that the project remains within the limits of the available human and physical resources. • Breaking large goals into achievable sub-tasks makes it easier to track progress and identify delays early.

Relevant (SMART)

Definition Goals must align with overall project aims and be clearly linked to stakeholder needs.

Explanation Relevance ensures that resources are not wasted on features that stakeholders do not actually want or need.

Scenario For a project building a school attendance system, a target to add "3D animated avatars" is rejected because it doesn't help track students.

Question (3-4 marks) Describe how a project manager ensures that project targets remain "relevant."

Mark Scheme: • By consulting with stakeholders to identify their core needs and requirements. • By regularly reviewing goals against the original project scope and objectives. • By rejecting feature requests that do not add direct value to the primary purpose of the system.

Time-bound (SMART)

Definition Goals must have clear and appropriate deadlines to avoid a lack of urgency or team burnout.

Explanation Time-bounding creates a sense of urgency and allows for the creation of a realistic schedule using tools like Critical Path Analysis.

Scenario A software team is given a deadline of April 30th to complete the Beta version of a new app.

Question (3-4 marks) State two consequences of failing to set "time-bound" targets in a systems development project.

Mark Scheme: • The project may suffer from a lack of urgency, leading to unnecessary delays in completion. • It becomes difficult to manage "time to market," which could lead to a loss of competitive advantage. • Without deadlines, it is impossible to accurately schedule dependent tasks or identify the critical path.

Risk Analysis

Definition Using a numbered risk matrix (e.g., 3x3 or 5x5) to identify potential risks and assign scores.

Explanation It works by quantifying risks based on their likelihood and their potential impact on the project, allowing managers to prioritize which risks to mitigate first.

Scenario An IT manager uses a 5x5 matrix to evaluate the risk of a cyber attack during a system upgrade.

Question (3-4 marks) Explain the process of conducting a risk analysis for an IT project.

Mark Scheme: • Identify potential risks such as technical issues, human error, or changing requirements. • Assign a score for the probability (likelihood) and the impact (severity) of each risk. • Calculate the total Risk Score (Probability x Impact) to prioritize which risks need immediate mitigation plans.

Risk Score

Definition Probability × Impact.

Explanation The risk score provides a numerical value that classifies the level of risk (e.g., 1-4 is "very low," 17-25 is "severe") to guide decision-making.

Scenario A project risk is identified as having a probability of 4 (Likely) and an impact of 3 (Significant).

Question (3-4 marks) Calculate the Risk Score for the scenario above and identify its risk level based on standard interpretations.

Mark Scheme: • Calculation: $4 \times 3 = 12$. • Identification: A score of 12 falls into the "Adequate" (10-16) risk level category. • Action: The manager should monitor this risk and ensure a basic mitigation plan is in place.

Functional Requirements

Definition Exactly what the user(s) should be able to do in the IT system (specific functions and features).

Explanation These are the "what" of the system, defining the specific tasks and operations the software must perform to be useful.

Scenario A library system must allow users to search for books by ISBN, title, or author.

Question (3-4 marks) Identify two functional requirements for a new e-commerce website.

Mark Scheme: • The system must allow users to add items to a digital shopping cart. • The system must provide a secure login feature for registered customers. • The system must generate an automated email confirmation after a successful purchase.

Non-functional Requirements

Definition How the system or product should perform (quality attributes, constraints, and characteristics).

Explanation These are the "how" of the system, focusing on performance, security, and usability rather than specific tasks.

Scenario A mobile app must load within 2 seconds and have an interface that is accessible for visually impaired users.

Question (3-4 marks) Explain the difference between a functional and a non-functional requirement using an example for each.

Mark Scheme: • A functional requirement describes a specific task the system must do, e.g., "the user must be able to reset their password". • A non-functional requirement describes a performance characteristic, e.g., "the system must be available 99.9% of the time". • Functional is about features; non-functional is about quality and performance attributes.

Critical Path Analysis (CPA)

Definition A method used to identify the sequence of tasks that must be completed on time to ensure the entire project meets the specified deadline.

Explanation CPA identifies "critical tasks" (those with zero slack) where any delay will immediately delay the entire project's completion date.

Scenario A project manager uses a network diagram to find the shortest possible time to complete a server installation.

Question (3-4 marks) Explain why identifying the critical path is important for a project manager.

Mark Scheme: • It highlights tasks that have zero slack, meaning they must be finished exactly on time. • It allows the manager to focus resources on the most important tasks to avoid project delays. • It helps in identifying non-critical tasks where there is "slack" or flexibility in the start time.

Slack

Definition The difference between the Latest Start (LS) and Early Start (ES).

Explanation Slack indicates the amount of time a non-critical task can be delayed without affecting the project's overall finish date.

Scenario Task B has an Early Start of Day 3 and a Latest Start of Day 5.

Question (3-4 marks) Calculate the slack for Task B and explain what this value means for the project manager.

Mark Scheme: • Calculation: $5 - 3 = 2$ days of slack. • Meaning: Task B can be delayed by up to 2 days without delaying the overall project. • Application: The manager could potentially move resources from Task B to a critical path task if needed.

Early Start (ES)

Definition The earliest possible date a task can begin, considering predecessors.

Explanation ES is determined by the finish times of all prerequisite tasks; a task cannot start until everything it depends on is complete.

Scenario Task C depends on Task A (finishes Day 2) and Task B (finishes Day 4).

Question (3-4 marks) Identify the Early Start for Task C based on the scenario and justify your answer.

Mark Scheme: • The Early Start for Task C is Day 4. • Justification: A task cannot start until all its predecessors are finished. • Since Task B finishes on Day 4, Task C must wait until that time even though Task A finished earlier.

Early Finish (EF)

Definition The earliest possible finish date ($EF = ES + D$).

Explanation EF represents the quickest a task can be completed if it starts as soon as possible and takes exactly its estimated duration.

Scenario Task D has an Early Start of Day 10 and an estimated duration of 5 days.

Question (3-4 marks) Calculate the Early Finish for Task D and state the formula used.

Mark Scheme: • Calculation: $10 + 5 = 15$. • Formula: $\text{Early Finish} = \text{Early Start} + \text{Duration}$. • The result (15) is the earliest possible day the task can be completed.

Late Start (LS)

Definition The latest possible start date without delaying project completion.

Explanation LS is calculated by working backward from the project deadline, ensuring that delayed starts for non-critical tasks don't push the final date.

Scenario A project must finish by Day 20; Task G (duration 4 days) must be finished for the project to end.

Question (3-4 marks) Explain how a project manager calculates the Late Start for a task.

Mark Scheme: • The manager performs a "backward pass" from the project's required finish date.
• They identify the Late Finish (LF) of the task based on the following tasks' late starts. • Formula:
 $\text{Late Start} = \text{Late Finish} - \text{Duration}$.

Late Finish (LF)

Definition The latest possible finish date ($\text{LF} = \text{LS} + \text{D}$).

Explanation LF identifies the final deadline for an individual task; missing this date will cause a cascading delay through the rest of the project.

Scenario Task H has a Late Start of Day 12 and a duration of 3 days.

Question (3-4 marks) Calculate the Late Finish for Task H.

Mark Scheme: • Calculation: $12 + 3 = 15$. • This means Day 15 is the latest the task can end without affecting the final project deadline. • If Task H ends on Day 16, the entire project will be delayed by one day.

Precedence Table

Definition A visual tool for scheduling and sequencing tasks, showing task durations, dependencies, and timing constraints.

Explanation It serves as the data source for creating Critical Path diagrams, clearly listing which tasks must happen before others.

Scenario A table lists Task A (no predecessor), Task B (predecessor A), and Task C (predecessor A).

Question (3-4 marks) Explain how a precedence table helps in the planning phase of a project.

Mark Scheme: • It clearly identifies dependencies, showing which tasks are prerequisites for others. • It provides estimated durations, allowing for the calculation of total project time. • It allows for the calculation of ES, EF, LS, LF, and Slack for every task.

Waterfall Method

Definition A sequential, linear approach to development where the project is divided into distinct, self-contained phases.

Explanation Waterfall works best when requirements are very clearly defined at the start and unlikely to change, as it is difficult to go back to previous phases.

Scenario A government department uses Waterfall to build a large-scale database for national census records.

Question (3-4 marks) Discuss one advantage and one disadvantage of using the Waterfall method for a large IT project.

Mark Scheme: • Advantage: It is highly structured and easy to manage because each phase has specific deliverables and a clear end point. • Disadvantage: It is inflexible; if requirements change during the implementation phase, it is very costly to go back to the analysis phase. • Disadvantage: The final product is only seen at the end of the project, so errors found late are difficult to fix.

Requirements Gathering

Definition Identification of stakeholders and consulting management and potential users on functionality.

Explanation This is the most critical first step in Waterfall, as any omissions here will negatively affect the design and implementation phases later.

Scenario A developer is using interviews and questionnaires to find out what users want from a new fitness tracking app.

Question (3-4 marks) Describe two methods of requirements gathering and provide a constraint for each.

Mark Scheme: • Interview: Direct interaction allows for non-verbal cues but is very time-consuming for large groups. • Questionnaire: Can gather feedback from hundreds of users quickly, but often suffers from low response rates. • Observation: Shows how users work in real-life but can be disruptive or obtrusive.

Interview

Definition Direct interaction; picks up non-verbal cues.

Explanation Interviews allow for in-depth, qualitative data gathering where the developer can ask follow-up questions to clear up ambiguity.

Scenario A system analyst meets one-on-one with the Head of Finance to discuss the requirements for a new payroll system.

Question (3-4 marks) Explain why an interview is a useful method for requirements gathering.

Mark Scheme: • It allows the analyst to pick up on non-verbal cues and gauge the stakeholder's level of concern or enthusiasm. • It allows for complex questions and immediate clarification of ambiguous requirements. • It builds a relationship between the developer and key decision-makers.

Observation

Definition Watching stakeholders in real-life environments.

Explanation Observation reveals how systems are *actually* used, which may differ significantly from how managers *think* they are used.

Scenario An IT consultant spends a day in a busy call centre watching how staff navigate their current customer database.

Question (3-4 marks) State one advantage and one constraint of using observation as a requirements gathering method.

Mark Scheme: • Advantage: It provides a realistic view of how a system is used in a high-pressure, real-world environment. • Constraint: It can be obtrusive or disruptive to the workers being watched. • Constraint: People may act differently when they know they are being observed (Hawthorne effect).

Document Analysis

Definition Reviewing existing reports/manuals for context.

Explanation This method provides a factual foundation of the current system's rules, data structures, and legal requirements.

Scenario A team reviews the current company policy manuals and annual error reports before designing a replacement IT system.

Question (3-4 marks) Explain how document analysis can help a developer understand the non-functional requirements of a system.

Mark Scheme: • Reviewing error logs can identify existing performance bottlenecks or reliability issues that must be improved. • Analyzing policy documents can reveal legal or security constraints the new system must follow. • It provides historical data on user patterns and system load.

Focus Group

Definition Collaborative discussions.

Explanation Focus groups encourage brainstorming and help identify conflicting requirements between different groups of users.

Scenario A developer gathers six teachers and three administrators to discuss how the new school grading portal should look.

Question (3-4 marks) Describe a potential constraint of using a focus group for requirements gathering.

Mark Scheme: • The discussion can be "hijacked" by dominant personalities, leading to a biased set of requirements. • It can be difficult and expensive to coordinate the schedules of multiple stakeholders for a single meeting. • Groupthink may occur, where participants agree with the majority rather than stating their true individual needs.

Questionnaire

Definition Structured feedback from large groups.

Explanation Questionnaires are excellent for gathering quantitative data (e.g., "70% of users want a mobile version") from a large, distributed user base.

Scenario An airline sends an online survey to 5,000 frequent flyers to ask which features they want in a new booking app.

Question (3-4 marks) Explain why a questionnaire is often used when a system has a very large number of external stakeholders.

Mark Scheme: • It is the most cost-effective and fastest way to reach a high volume of people. • It produces structured data that is easy to analyze using statistical tools. • It allows stakeholders to provide feedback at their own convenience, increasing the chance of participation.

System Design

Definition Expanding requirements into detailed technical and visual designs.

Explanation This phase creates the blueprint for the system, including UI layouts, database schemas, and information flow diagrams.

Scenario A designer creates a wireframe for a website's navigation menu and a data dictionary for the backend database.

Question (3-4 marks) Identify three components that would be created during the system design phase.

Mark Scheme: • User Interface (UI) designs or wireframes showing the layout of the screens. • A Data Dictionary defining the tables, fields, and validation rules for the database. • Information Flow Diagrams (IFD) showing how data travels through the system.

Implementation

Definition Translating designs into working code, building infrastructure, and creating databases.

Explanation This is the "execution" phase where the system is physically built based on the blueprints established in the design phase.

Scenario Programmers begin writing the Python code for a new e-commerce engine while engineers set up the cloud servers.

Question (3-4 marks) Describe two tasks that are typically performed during the implementation phase.

Mark Scheme: • Coding: Programmers translate the technical designs into a functional programming language. • Database Creation: Building the tables and relationships identified in the data dictionary. • Infrastructure Building: Installing hardware or configuring cloud environments for the software to run on.

Unit Testing

Definition Testing individual modules/units.

Explanation Unit testing is performed by developers during the coding process to ensure that small blocks of code function correctly in isolation.

Scenario A developer writes a small test script to ensure that the "calculate tax" function in a payroll app returns the correct value.

Question (3-4 marks) Explain why unit testing is the first type of testing performed in the development process.

Mark Scheme: • It identifies logic errors in small sections of code before they are combined with other modules. • It makes debugging easier because the developer knows exactly which module contains the error. • It ensures that the basic building blocks of the system are reliable before integration testing begins.

Integration Testing

Definition Testing how units communicate/share data when combined.

Explanation Integration testing focuses on the interfaces between modules, ensuring that data passed from one unit is correctly received and processed by the next.

Scenario The login module and the user profile module are linked together to ensure that the correct user data is loaded after a successful login.

Question (3-4 marks) Describe the primary goal of integration testing.

Mark Scheme: • To ensure that different software modules or units work correctly together when combined. • To verify that data is passed between modules accurately and in the correct format. • To identify issues in the communication protocols between different parts of the system.

System Testing

Definition Testing the entire system against requirements.

Explanation Performed by a dedicated QA team, this phase ensures the fully integrated system meets all functional and non-functional requirements.

Scenario The QA team runs the completed banking app on multiple devices to check if it meets the 2-second load time requirement.

Question (3-4 marks) Explain how system testing differs from unit testing.

Mark Scheme: • Unit testing focuses on isolated modules, whereas system testing evaluates the entire completed product. • Unit testing is usually done by the programmer, while system testing is often done by an independent QA team. • System testing checks the product against the original user requirements, not just the code logic.

Acceptance Testing

Definition Final confirmation by stakeholders that the system is ready for deployment.

Explanation This is the "final sign-off" where actual users test the system in a real-world scenario to ensure it meets their practical needs.

Scenario The Head of Sales logs into the new CRM system and confirms that it can generate all the required monthly reports accurately.

Question (3-4 marks) Discuss why acceptance testing is a critical milestone in the Waterfall methodology.

Mark Scheme: • It is the final opportunity for the client to verify that the system delivered matches their original requirements. • It serves as a legal or contractual "sign-off," moving the project from the developer to the maintenance phase. • It identifies any usability issues that might not have been caught by the technical QA team.

Deployment

Definition Migrating the system to target devices/locations.

Explanation Deployment involves the physical installation of software, configuration of networks, and the training of end-users on the new system.

Scenario Engineers visit every branch of a bank to install new ATM software and train the staff on the new interface.

Question (3-4 marks) Identify three activities that occur during the deployment phase of a project.

Mark Scheme: • Installation of physical hardware or configuration of cloud server instances. • Migration of data from the old system to the new one. • Training end-users to ensure they can use the new system effectively.

Maintenance

Definition Reactive process of fixing errors, updates, and streamlining existing systems.

Explanation Maintenance is the longest phase of the lifecycle, covering bug fixes (corrective), environment changes (adaptive), and upgrades (perfective).

Scenario A social media company releases a patch to fix a security vulnerability discovered two months after the app launched.

Question (3-4 marks) Explain why the maintenance phase is necessary after a system has been deployed.

Mark Scheme: • To fix bugs or logic errors (corrective maintenance) that were only discovered through mass user usage. • To update the system to work with new hardware or operating system versions (adaptive maintenance). • To add new features or improve performance based on changing business needs (perfective maintenance).

Agile

Definition A modern approach focused on small but continuous and rapid improvements to a system.

Explanation Agile is iterative and incremental, allowing teams to respond quickly to feedback and deliver a "working" system much earlier than Waterfall.

Scenario A startup building a new fitness app releases a version with just a step counter first, then adds a heart rate monitor two weeks later.

Question (3-4 marks) Discuss two benefits of using an Agile approach for a software development project.

Mark Scheme: • Faster "time to market" as a basic working system can be released to users very early. • Increased flexibility allows the team to adapt to changing user feedback after every iteration. • Improved product quality because testing and reviews are constant rather than just at the end.

Sprint

Definition A single iteration and incrementation cycle, typically lasting between 1 and 4 weeks.

Explanation Sprints are the heartbeat of Agile, providing a fixed timebox to complete a specific set of requirements from the project plan.

Scenario The development team sets a goal to complete the "User Login" and "Password Reset" features by the end of a 14-day cycle.

Question (3-4 marks) Describe the activities that occur during a single Sprint in Agile development.

Mark Scheme: • Planning: The team identifies specific requirements to focus on for this cycle. • Develop/Test: The team codes the features and performs immediate testing and reviews. • Release/Review: The new increment is shown to stakeholders for feedback before the next sprint starts.

Iterative

Definition The project continually reviews and improves features and quality.

Explanation Iteration means going over the same process multiple times; each "loop" refines the product based on what was learned in the previous one.

Scenario A team builds a basic website layout, shows it to the client, realizes the colors are wrong, and redraws it in the next cycle.

Question (3-4 marks) Explain how "iterative" development helps improve the quality of an IT system.

Mark Scheme: • It allows for continuous feedback from users, ensuring the final product matches their needs. • Design flaws are identified and corrected in the next iteration rather than being baked into the final product. • It allows the team to experiment with different solutions and keep only the ones that work.

Incremental

Definition Small, measurable developments and changes that move the project closer to completion.

Explanation Incrementation is about adding "pieces" to the system; each sprint delivers a functional addition (e.g., adding a payment gateway to an existing cart).

Scenario A video game team completes the "character movement" in week one, then adds "combat" in week two.

Question (3-4 marks) Contrast the concepts of "iterative" and "incremental" development.

Mark Scheme: • Iterative is about refinement and repeated cycles to improve existing work. • Incremental is about addition and building the system piece-by-piece until it is complete. • Iterative improves the "how"; incremental adds the "what".

Scrum

Definition A popular framework involving four key activities.

Explanation Scrum provides a specific structure for managing Agile projects, emphasizing team collaboration and short, time-boxed cycles.

Scenario A team uses a "Scrum Master" to facilitate meetings and help remove any obstacles blocking the developers' progress.

Question (3-4 marks) Identify and describe two of the key activities associated with the Scrum framework.

Mark Scheme: • Sprint Planning: Setting the goals and specific tasks for the upcoming 1-4 week cycle. • Daily Scrum: Short meetings to share progress, align the team, and resolve any immediate problems. • Sprint Review/Retrospective: Meetings at the end of a sprint to demonstrate progress and identify areas for improvement.

Sprint Planning

Definition Defining goals and tasks for the upcoming sprint.

Explanation This meeting ensures the team only takes on an "achievable" amount of work for the cycle, preventing over-commitment.

Scenario On Monday morning, the team looks at the "backlog" of tasks and decides which three features to build this week.

Question (3-4 marks) Explain why sprint planning is necessary for a successful Agile project.

Mark Scheme: • It ensures the team is focused on a clear, specific set of targets for the duration of the sprint. • It allows the team to check that the selected tasks are "achievable" within the 1-4 week timeframe. • It identifies any prerequisites or dependencies that might block development during the sprint.

Daily Scrum

Definition Short daily meetings to share progress and resolve problems.

Explanation Also called a "stand-up," this meeting maintains high communication and ensures that any "bottlenecks" are identified within 24 hours.

Scenario The team meets for 15 minutes every morning to answer: What did I do yesterday? What will I do today? Is anything blocking me?

Question (3-4 marks) Describe the role of the Daily Scrum in risk mitigation.

Mark Scheme: • It allows team members to flag "blockers" or technical issues immediately so the manager can take action. • it ensures that the team remains coordinated and that no two

members are duplicating effort. • It provides a daily "monitoring" phase to ensure the sprint is still on track for its deadline.

Sprint Review

Definition Collecting stakeholder feedback at the end of a sprint.

Explanation The review is external-facing; it is where the team demonstrates "working software" to stakeholders to confirm they are moving in the right direction.

Scenario At the end of the month, the developers show the client the new "checkout" page and ask for their opinion on the button layout.

Question (3-4 marks) Explain how a Sprint Review contributes to the "Relevant" part of SMART targets.

Mark Scheme: • It allows stakeholders to confirm that the new features align with their actual business needs. • If a feature is not relevant or user-friendly, the feedback can be used to change course in the next sprint. • It ensures that resources are not wasted on developing unwanted functionality.

Sprint Retrospective

Definition Reflecting on the past sprint to identify future improvements.

Explanation The retrospective is internal-facing; the team discusses how they worked together and what technical or process changes could make the *next* sprint better.

Scenario After a difficult sprint, the team realizes they didn't have enough test data and agrees to build a test generator for next time.

Question (3-4 marks) Discuss the difference between a Sprint Review and a Sprint Retrospective.

Mark Scheme: • A Sprint Review is about the *product* (features/stakeholder feedback), while a Retrospective is about the *process* (how the team worked). • The Review involves stakeholders;

the Retrospective is typically just for the internal development team. • Review focuses on "what" was built; Retrospective focuses on "how" it was built.

Kanban

Definition Focuses on continuous flow and visualising tasks on a board.

Explanation Kanban uses columns like "To Do," "In Progress," and "Done" to limit work-in-progress and identify where tasks are getting stuck.

Scenario A helpdesk team uses a digital board with sticky notes to track every open support ticket until it is resolved.

Question (3-4 marks) Describe one way that using a Kanban board improves project visibility.

Mark Scheme: • It provides a real-time visual representation of the project's current status for all stakeholders. • It highlights "bottlenecks" where too many tasks are piling up in the "In Progress" column. • It ensures that team members always know which task is the highest priority to work on next.

Lean

Definition Adapts principles from lean manufacturing to IT development.

Explanation Lean focuses on "eliminating waste," which in IT means removing unnecessary meetings, redundant code, and features that don't add value.

Scenario A development team decides to stop writing 50-page technical manuals that nobody reads and instead creates short video tutorials.

Question (3-4 marks) Explain the core philosophy of the "Lean" approach in systems development.

Mark Scheme: • The primary goal is the "elimination of waste" to improve efficiency. • It emphasizes delivering value to the customer as quickly as possible with minimal unnecessary steps. • It encourages continuous improvement and empowering the team to make decisions.

Information Flow Diagram (IFD)

Definition Visual representations showing how information travels within and between systems, organisations, and processes.

Explanation IFDs map the "journey" of data, identifying where it comes from (entities), how it is handled (processes), and where it ends up.

Scenario An analyst draws a diagram to show how a customer's order moves from the website to the warehouse and then to the shipping company.

Question (3-4 marks) Identify the four main components of an Information Flow Diagram.

Mark Scheme: • Entities (Sources/Destinations of information). • Flows (Direction of data travel). • Processes (Transformations of data). • Decisions (Points where data path depends on conditions).

Entity (IFD)

Definition Sources or destinations of information (customers, databases, external systems); represented by rectangles.

Explanation Entities are the external origins or final landing spots for data; they do not process the data themselves within the scope of that diagram.

Scenario In a library IFD, "Student" is an entity that provides a "Library Card Number" and receives "Book Availability."

Question (3-4 marks) Explain why a "Database" is often represented as an entity in an Information Flow Diagram.

Mark Scheme: • It acts as a permanent storage location (destination) for information after it has been processed. • It acts as a source of information that is needed by other processes in the system. • Representing it as a rectangle (entity) distinguishes it from the actual "searching" or "saving" actions (processes).

Flow (IFD)

Definition Direction of information flow, shown using arrows.

Explanation Flows illustrate the movement of specific data items (e.g., "Invoice," "Error Message") between entities and processes.

Scenario An arrow labeled "Payment Confirmation" points from the "Bank System" entity to the "Generate Receipt" process.

Question (3-4 marks) Describe the importance of labeling "Flow" arrows in an IFD.

Mark Scheme: • Labels identify exactly *what* information is being moved (e.g., "Order ID" vs "Customer Name"). • The arrow head shows the "direction" of travel, clarifying which component is the source and which is the recipient. • Without labels, the diagram shows connections but fails to explain the logic of the system.

Process (IFD)

Definition Transformations or processing of information; represented by boxes.

Explanation Processes are the "verbs" of the diagram—they take incoming data and change it into a new form of information.

Scenario A box labeled "Calculate Total" takes "Item Price" and "Quantity" as inputs and produces "Grand Total."

Question (3-4 marks) Contrast an Entity and a Process in an Information Flow Diagram.

Mark Scheme: • An Entity is a source or destination (noun) represented by a rectangle, while a Process is an action (verb) represented by a box. • Processes actively "transform" or change data, whereas Entities simply send or receive it. • Entities are often external to the immediate software system, while Processes happen within it.

Decision (IFD)

Definition Logical decisions where flow depends on specific conditions; represented by diamonds.

Explanation Decisions create branching paths in the data flow, such as "If payment valid, go to process A; if invalid, go to process B".

Scenario A diamond asks "In Stock?" and sends data to "Pack Order" if True, or "Backorder Item" if False.

Question (3-4 marks) Explain how a "Decision" component adds logical complexity to an Information Flow Diagram.

Mark Scheme: • It introduces "branching," allowing for multiple possible outcomes based on specific conditions. • It allows the diagram to show error handling paths (e.g., "invalid password") rather than just a single linear flow. • It identifies points where the system must evaluate data before choosing the next process.

Level 0 Diagram

Definition High-level overview of the entire system and external interactions.

Explanation Also known as a Context Diagram, it shows the system as a single process and focuses on the external entities it communicates with.

Scenario A diagram shows one large box labeled "Banking System" connected to "Customer," "ATM," and "Tax Office."

Question (3-4 marks) Describe the purpose of a Level 0 Information Flow Diagram.

Mark Scheme: • To provide a high-level "context" of the entire system for stakeholders without technical detail. • To identify all external entities that interact with the system. • To show the primary inputs and outputs that cross the system boundary.

Level 1 Diagram

Definition More detailed view focusing on major subsystems and flows within them.

Explanation A Level 1 diagram "decomposes" the single box of Level 0 into its major internal components, such as "Login," "Search," and "Payment".

Scenario The "Banking System" box is broken down to show how data moves between "Validation," "Transaction," and "Database" subsystems.

Question (3-4 marks) Explain why a developer would create a Level 1 diagram after finishing a Level 0 diagram.

Mark Scheme: • To provide more "detail" about the internal workings and major processes of the system. • To show how information flows between the different internal subsystems. • To help developers identify the specific functional requirements that need to be coded.

Level 2 Diagram

Definition Highly detailed view exploring specific processes and handling of information.

Explanation Level 2 diagrams provide the most granular view, mapping out every single step and decision within a single process from the Level 1 diagram.

Scenario A diagram focuses solely on the "Validate Password" process, showing every check and error message possible.

Question (3-4 marks) State which stakeholders would find a Level 2 IFD most useful and justify your answer.

Mark Scheme: • Software Developers/Programmers. • Justification: It provides the "highly detailed" logic required to write the actual code and handle specific data transformations. • It allows for precise "unit testing" planning by showing every possible logical branch.

HOW TO ANSWER GUIDE

1. DEFINE (1-2 marks)

- **What to include:** The exact technical terminology from the textbook. Use key "anchors" (e.g., for *Agile*, mention "iterative and incremental").
- **What to avoid:** Vague language like "it's when you do..." or circular definitions (e.g., "Project Management is managing a project").
- **Pro Tip:** If you forget the exact wording, describe its *primary purpose*.

2. EXPLAIN (3-4 marks)

- **Structure:** [Definition/Identification] + [Mechanism (how/why it works)] + [Benefit/Impact].
- **Keywords:** Use connective words like "therefore," "resulting in," or "which ensures."
- **Example:** "Critical Path Analysis identifies the longest sequence of tasks (1). This allows a manager to see which tasks have zero slack (2), ensuring that resources are prioritized to avoid any delay to the final deadline (3)."

3. DISCUSS/EVALUATE (6-8 marks)

- **Structure:** Introduction -> Point 1 (Advantage + Evidence) -> Point 2 (Disadvantage + Evidence) -> Scenario Application -> Conclusion.
- **Scenario Use:** You *must* link your points back to the specific business in the question. If the question is about a bank, talk about data security and uptime, not generic "safety."
- **What the examiner wants:** Balanced judgment. Don't just list facts; weigh them. "While Agile is faster (Adv), the uncertain timelines might be unacceptable for a government project with a fixed budget (Evaluation)."

4. Common mistakes on Topic 15

- **Confusing Iterative vs. Incremental:** Remember: Iterative = *polishing* the same piece; Incremental = *adding* a new piece.
- **Mixing up Review vs. Retrospective:** Review is about the *product* (with the client); Retrospective is about the *team/process* (internal).
- **Slack Calculation Errors:** Always remember $\$Slack = LS - ES$. Students often mix up Late Start and Late Finish.
- **Waterfall Inflexibility:** Students often say Waterfall has "no testing." It *does* have testing, but it happens very late in the process, which is the actual problem.
- **IFD Symbols:** Using a box (Process) for a person (Entity) is a common mark-loser. Remember: Rectangles = People/Systems; Boxes = Actions.

TOPIC 16

MACHINE LEARNING

Definition A branch of artificial intelligence (AI) that allows computers to learn and improve without being explicitly programmed.

Explanation It works by identifying trends and correlations within large datasets and applying that "learning" to new, unseen data rather than following fixed, pre-determined rules.

Scenario A music streaming service uses a system to suggest new songs to users based on their previous listening history and the habits of similar listeners.

Question (3-4 marks) Explain how machine learning differs from a traditional expert system in the context of song recommendations.

Mark Scheme:

- Machine learning identifies trends and correlations in datasets rather than following fixed rules.
- It improves automatically over time as more data is processed.
- An expert system would rely on a static knowledge base of rules pre-programmed by humans.

ALGORITHM

Definition Sets of instructions used to process data and make decisions.

Explanation In machine learning, algorithms use mathematical and statistical processes to analyze data and improve the model's performance over time.

Scenario A social media platform uses a set of instructions to determine which posts should appear at the top of a user's feed.

Question (3-4 marks) Describe the role of an algorithm in a machine learning model.

Mark Scheme:

- Provides the set of instructions for processing input data.
- Uses statistical processes to identify patterns.

- Adjusts internal parameters to reduce error rates during the learning process.
-

TRAINING SET

Definition The larger portion of data used to train the model.

Explanation The model uses this data to learn patterns and adjust its internal parameters to minimize error rates.

Scenario A developer provides a system with 10,000 images of various fruits, all labeled correctly, so the system can learn to identify them.

Question (3-4 marks) Explain the importance of the training set in the development of an image recognition system.

Mark Scheme:

- Allows the model to learn the specific characteristics/patterns of the objects (e.g., fruit shapes/colors).
 - Used to adjust internal parameters to ensure the model makes fewer mistakes.
 - The quality and quantity of this set directly impact the growth and accuracy of the AI.
-

TEST SET

Definition A smaller portion of data the model is not exposed to during training.

Explanation It is used to evaluate the quality and accuracy of the model by testing it on data it hasn't "seen" before.

Scenario After learning to identify fruits from a labeled batch, the system is given 500 new, unlabeled fruit images to see if it can correctly categorize them.

Question (3-4 marks) Distinguish between a training set and a test set.

Mark Scheme:

- The training set is the larger portion used for initial learning and parameter adjustment.
 - The test set is data the model was not exposed to during the training phase.
 - The test set is used specifically for evaluating the final accuracy and quality of the predictions.
-

SUPERVISED LEARNING

Definition Uses a "labelled" dataset where each training example contains an input and a corresponding output (actual category).

Explanation The goal is to determine the relationship or "mapping function" between the data and the label so the model can predict labels for new data.

Scenario An email filter is trained using a dataset where thousands of messages are already marked as either "Spam" or "Inbox."

Question (3-4 marks) Explain how supervised learning is used to create a functional email spam filter.

Mark Scheme:

- Uses a labeled dataset where inputs (emails) are linked to specific outputs (spam/not spam).
 - The system identifies the relationship/patterns between the email content and the labels.
 - It then applies this mapping function to categorize new, incoming emails.
-

UNSUPERVISED LEARNING

Definition Uses "unlabelled" or "unknown" data where the model finds hidden patterns or structures without prior knowledge.

Explanation This method is used to discover the inherent structure of a dataset without influence from human-provided labels or bias.

Scenario A marketing firm provides a machine learning model with raw customer purchase data to find groups of people with similar buying habits without pre-defining the groups.

Question (3-4 marks) Describe one advantage of using unsupervised learning over supervised learning.

Mark Scheme:

- It can find hidden patterns or structures that humans might not be aware of.
 - It operates without the need for pre-labeled data, which can be time-consuming to produce.
 - It avoids human bias because there are no pre-determined categories.
-

REGRESSION

Definition A type of supervised learning used to predict continuous outcomes.

Explanation It establishes linear relationships between variables to forecast future values based on historical trends.

Scenario A real estate application predicts the future market price of a house based on its size, location, and previous sale prices.

Question (3-4 marks) Identify and explain the type of machine learning problem used to predict house prices.

Mark Scheme:

- Regression problem.
 - It is a type of supervised learning.
 - It predicts a continuous outcome (the specific price) by analyzing relationships in historical data.
-

CLASSIFICATION

Definition A type of supervised learning where the goal is to predict categorical class labels based on past observations.

Explanation It works by comparing new data points against known, labeled categories to assign the most appropriate label.

Scenario A medical system analyzes a patient's scan to determine if a tumor is "benign" or "malignant."

Question (3-4 marks) Explain how a classification algorithm would determine the status of a medical scan.

Mark Scheme:

- Uses past observations/labeled data of benign and malignant tumors.
 - Assigns a categorical label to the new data point (the scan).
 - This is an application of supervised learning.
-

CLUSTERING

Definition An unsupervised learning technique used to group similar data points based on certain characteristics.

Explanation The system analyzes unlabelled data to find natural groupings or "clusters" based on similarities in the dataset.

Scenario An online bookstore groups its millions of books into clusters based on shared themes and writing styles without using existing genre labels.

Question (3-4 marks) Explain why clustering is considered an unsupervised learning technique.

Mark Scheme:

- It processes "unlabelled" data where the categories are not known in advance.
 - The model finds the structure/groupings on its own without prior human knowledge.
 - It does not use a mapping function between inputs and pre-defined outputs.
-

ANOMALY DETECTION

Definition Identify data points falling outside "normal" behavior.

Explanation Used in both supervised and unsupervised models to spot rare items or events that raise suspicions by differing significantly from the rest of the data.

Scenario A credit card company's system flags a high-value purchase made in a foreign country as "suspicious" because it doesn't match the customer's usual spending profile.

Question (3-4 marks) Describe how anomaly detection helps protect bank customers.

Mark Scheme:

- Establishes a baseline of "normal" behavior based on past transactions.
 - Identifies specific data points (transactions) that fall outside this normal behavior.
 - Allows for immediate intervention (e.g., freezing the card) to prevent fraud.
-

NATURAL LANGUAGE PROCESSING (NLP)

Definition Focuses on interaction between computers and humans using regular everyday language.

Explanation It allows systems to understand user intent, accents, and regional variations to provide more human-like responses.

Scenario A customer types a question into a retail website's chat box, and the system understands the intent and provides a relevant answer.

Question (3-4 marks) Explain two benefits of using NLP in customer service chatbots.

Mark Scheme:

- Improves responsiveness/efficiency by handling queries without human intervention.
 - Understands "everyday language" and user intent rather than requiring specific keywords.
 - Can handle regional variations and accents if paired with speech recognition.
-

SPEECH RECOGNITION

Definition Processing spoken language into a form that a computer can understand.

Explanation Modern systems use machine learning and context to improve the accuracy of how audio signals are converted into text or commands.

Scenario A driver dictates a text message while driving, and the car's infotainment system converts the voice into text.

Question (3-4 marks) Explain how speech recognition serves as an accessibility tool.

Mark Scheme:

- Provides a distraction-free/hands-free method of interaction for users (e.g., while driving).
 - Allows individuals with limited motor functions or sight loss to control computer systems.
 - Modern systems use context and machine learning to accurately process different accents.
-

IMAGE RECOGNITION

Definition Harnesses machine learning to identify objects within images.

Explanation It identifies patterns and features in visual data to categorize objects, even when there are significant variations in lighting or angle.

Scenario A self-driving car identifies a pedestrian crossing the road and applies the brakes.

Question (3-4 marks) Explain why image recognition is computationally difficult.

Mark Scheme:

- There are significant variations in how objects appear (angles, lighting, distance).
 - Requires analyzing vast amounts of pixel data to find identifying patterns.
 - Relies on extensive training sets to achieve high accuracy for safety-critical tasks (like self-driving).
-

PATTERN RECOGNITION

Definition Finding trends, correlations, and similarities in datasets.

Explanation It allows computers to discover and adapt to patterns in data that are beyond human capabilities to detect manually.

Scenario A weather monitoring system analyzes historical atmospheric data to identify recurring patterns that precede major storms.

Question (3-4 marks) State how pattern recognition is used in the field of Big Data analytics.

Mark Scheme:

- Identifies trends and correlations within very large/complex datasets.
 - Discovers similarities that allow for predictive outcomes (e.g., forecasting).
 - Adapts to new patterns as the data evolves without explicit re-programming.
-

VIRTUAL REALITY

Definition Provides an immersive experience within a computer-generated environment to create the feeling that the user is actually present.

Explanation It combines high-resolution visuals, spatial audio, and sensory feedback to simulate a realistic or fantasy world.

Scenario A trainee surgeon practices a complex heart procedure using a simulated 3D environment before performing it on a real patient.

Question (3-4 marks) Explain one benefit of using virtual reality for professional training.

Mark Scheme:

- Provides a safe environment to practice hazardous or high-stakes tasks.
 - Immersion creates a feeling of "actual presence," making the simulation more effective.
 - Allows for repetitive practice of rare scenarios without the cost of physical resources.
-

STEREOSCOPIC SCREENS

Definition Combining two images taken at slightly different angles to provide a three-dimensional view.

Explanation This technology mimics human binocular vision to create a sense of depth and perspective in a virtual environment.

Scenario A VR headset displays two slightly offset images of a virtual forest to make the trees appear at different distances from the user.

Question (3-4 marks) Describe how stereoscopic screens create a sense of immersion in VR.

Mark Scheme:

- They display two images simultaneously, each taken at a slightly different angle.
 - This provides a three-dimensional view to the user.
 - It creates a realistic sense of depth, making virtual objects appear to have physical volume.
-

HAPTIC FEEDBACK

Definition Simulated touch used to enhance immersion.

Explanation It uses vibrations or motors to create physical sensations that correspond to actions taking place in the virtual world.

Scenario While playing a VR game, a user feels a distinct vibration in their handheld controller when their virtual sword hits a shield.

Question (3-4 marks) Explain the role of haptic feedback in a VR system.

Mark Scheme:

- Engages the user's sense of touch to create realism.
 - Provides physical confirmation (feedback) of interactions within the computer-generated environment.
 - Enhances immersion by making the virtual world feel more tangible.
-

HEADSET

Definition Goggle-like devices using stereoscopic screens... often including headphones and head-tracking technology.

Explanation The headset is the primary hardware interface that blocks out the physical world and replaces it with computer-generated sensory input.

Scenario A real estate agent gives a client a device to wear that allows them to walk through a virtual model of a house that hasn't been built yet.

Question (3-4 marks) Identify two pieces of technology integrated into a modern VR headset.

Mark Scheme:

- Stereoscopic screens to provide 3D visuals and depth.
- Head-tracking technology to adjust the view based on the user's movement.
- (Accept: Headphones for spatial audio/immersion).

CONTROLLERS

Definition Handheld devices used to interact with the virtual world and track user movements.

Explanation They allow the user to manipulate virtual objects and provide a bridge between physical gestures and digital actions.

Scenario A user moves their arm in the real world while holding a device, and their virtual hand picks up a glass in the simulation.

Question (3-4 marks) Describe the function of controllers in a VR environment.

Mark Scheme:

- Used to track the user's hand/arm movements in real-time.
- Allow the user to interact with or manipulate virtual objects.
- Often provide haptic feedback (vibrations) to confirm interactions.

INTERNET OF THINGS

Definition A network of devices not traditionally considered "smart" (e.g., household appliances) that have been enabled to generate, send, and receive data.

Explanation It works by equipping everyday objects with sensors and connectivity to allow for remote monitoring and task automation.

Scenario A city installs a network of streetlights that only turn on when their sensors detect a pedestrian or vehicle nearby.

Question (3-4 marks) State two benefits of the Internet of Things (IoT).

Mark Scheme:

- Convenience through the automation of routine tasks.
- Increased efficiency, saving time, energy, or resources.
- (Accept: Safety via remote monitoring of sensors).

SENSORS

Definition Devices that monitor conditions and convert them into digital signals.

Explanation They are the foundational component of IoT, allowing "dumb" objects to collect data about their environment (e.g., temperature, motion, location).

Scenario A smart thermostat uses a thermistor to detect when the room temperature drops below 18 degrees Celsius.

Question (3-4 marks) Identify three types of sensors used in IoT devices and describe a use for one of them.

Mark Scheme:

- Types: GPS receiver, accelerometer, gyroscope, thermistor, microphone, etc..
 - Use: e.g., Thermistor to monitor temperature in a smart home/refrigerator.
 - Function: Sensors convert physical conditions into digital signals for processing.
-

EMBEDDED SYSTEMS

Definition Specialized, integrated computing devices (microprocessor, memory, I/O) designed for narrow, specific tasks.

Explanation They are designed to be compact, power-efficient, and highly reliable for dedicated functions within a larger mechanical or electrical system.

Scenario A modern washing machine contains a dedicated microprocessor that controls the water level, spin speed, and duration of different wash cycles.

Question (3-4 marks) Explain why embedded systems are used in IoT devices rather than general-purpose computers.

Mark Scheme:

- They are designed for a narrow, specific task, making them more efficient.
 - They are compact and can be integrated directly into small appliances.
 - They are power-efficient and highly reliable for continuous operation.
-

EDGE STORAGE

Definition Data processed/stored on the device itself or nearby gateways.

Explanation It is used to reduce network congestion and latency by handling data locally rather than sending everything to a central cloud server.

Scenario A security camera stores the last 24 hours of high-definition video on its internal SD card rather than uploading it all to the internet.

Question (3-4 marks) Explain one advantage of edge storage in an IoT security system.

Mark Scheme:

- Reduces network congestion by not requiring continuous high-bandwidth uploads.
- Decreases latency because data can be accessed or processed locally.
- Provides a level of redundancy if the internet connection is lost.

CLOUD STORAGE

Definition Centralized platforms offering scalability and accessibility.

Explanation It allows vast amounts of IoT data from millions of devices to be aggregated, analyzed, and accessed from anywhere via the internet.

Scenario A fitness tracker uploads a user's year-long heart rate history to a remote server so it can be viewed on any device.

Question (3-4 marks) Compare cloud storage with edge storage for an IoT application.

Mark Scheme:

- Cloud storage offers superior scalability for very large datasets.
- Cloud data is accessible from any location with an internet connection.
- Unlike edge storage, cloud storage relies on a stable network connection and may have higher latency.

FOG COMPUTING

Definition Data stored in distributed mini-data centres geographically closer to devices.

Explanation It acts as an intermediate layer between the "edge" and the "cloud" to provide local processing power for time-sensitive tasks.

Scenario A smart city uses small server boxes located at every major intersection to process traffic sensor data instantly before sending summaries to the main city server.

Question (3-4 marks) Describe how fog computing improves the performance of an IoT network.

Mark Scheme:

- Places mini-data centers geographically closer to the devices.
- Reduces the distance data must travel, lowering latency.

- Handles processing for time-sensitive tasks that cannot wait for a round-trip to the central cloud.
-

HYBRID STORAGE

Definition Combines edge, cloud, and fog to balance performance, security, and cost.

Explanation This approach allows an organization to keep sensitive or time-critical data local while utilizing the cloud for massive data analytics and long-term storage.

Scenario A manufacturing plant uses local storage for immediate machine alerts but sends weekly efficiency reports to a cloud-based dashboard.

Question (3-4 marks) Explain why an organization might choose a hybrid storage solution for its IoT data.

Mark Scheme:

- Balances performance (low latency via edge/fog) with scalability (cloud).
 - Optimizes costs by only sending necessary data to expensive cloud platforms.
 - Improves security by keeping sensitive data on-premises (edge) while using the cloud for general tasks.
-

STANDARDISATION

Definition Rules that ensure different devices/systems can communicate seamlessly.

Explanation In the IoT, standardization is essential because it allows devices from different manufacturers to work together using shared protocols.

Scenario A homeowner can control their smart lights, locks, and oven from a single app because all the devices follow the same communication protocol.

Question (3-4 marks) Explain why a lack of standardization is a challenge for the IoT industry.

Mark Scheme:

- Different manufacturers may use proprietary protocols, preventing devices from communicating.
 - Users may become "locked in" to a single brand to ensure their smart home works.
 - It complicates the development of universal software or security updates.
-

API (APPLICATION PROGRAMMING INTERFACE)

Definition Sets of rules and protocols that tell systems how to communicate and share data.

Explanation APIs act as a bridge that allows different software applications to interact with IoT hardware and access gathered data securely.

Scenario A weather app uses a protocol to pull live temperature data from a network of thousands of private smart thermometers.

Question (3-4 marks) Describe the role of an API in an IoT solution.

Mark Scheme:

- Provides a set of rules for how two systems should exchange data.
 - Allows different software to "talk" to IoT devices seamlessly.
 - Facilitates standardization across diverse hardware platforms.
-

PRIVACY CONCERNS

Definition Risks related to the collection of personal data (location, health metrics, behavior).

Explanation Because IoT devices often monitor intimate details of daily life, unauthorized access to this data can lead to identity theft or personal safety risks.

Scenario A smart speaker accidentally records a private medical conversation and uploads the audio to a server where it is accessed by an unauthorized employee.

Question (3-4 marks) Discuss one method an IoT manufacturer can use to address privacy concerns.

Mark Scheme:

- Implement strong encryption to ensure data is unreadable to unauthorized parties.
 - Use robust access controls so data is seen only by those with a valid need.
 - Provide clear transparency to users about what data is collected and how it is used.
-

DATA INTEGRITY

Definition Ensuring data remains free from error and useable from collection through use to archiving.

Explanation In an IoT context, maintaining integrity is critical because tampered or incorrect data (e.g., false sensor readings) can lead to catastrophic failures in industrial or medical systems.

Scenario An attacker changes the data packets from a temperature sensor in a nuclear reactor to show "normal" even though the core is overheating.

Question (3-4 marks) Explain why data integrity is vital for industrial IoT systems.

Mark Scheme:

- Decisions (often automated) are based on the reliability of sensor data.
 - Tampering with data can have "catastrophic effects" on safety-critical systems.
 - Validation is required to ensure data stays within expected, safe parameters.
-

UNAUTHORISED ACCESS

Definition Threat actors exploit vulnerabilities (default passwords, unpatched software) for data theft or espionage.

Explanation IoT devices are often "weak links" in a network because they may lack the complex security features found on traditional computers.

Scenario A hacker gains access to a corporate network by guessing the simple default password on a smart coffee machine in the office breakroom.

Question (3-4 marks) Identify two vulnerabilities that lead to unauthorized access in IoT devices.

Mark Scheme:

- The use of simple or "default passwords" that are never changed.
 - Unpatched or "outdated software" that contains known security flaws.
 - (Accept: Use of insecure communication channels like unencrypted Wi-Fi).
-

DENIAL OF SERVICE (DoS) ATTACK

Definition Overwhelming networks/devices with traffic to disrupt operations.

Explanation IoT systems are particularly vulnerable because they rely heavily on constant communication channels; a DoS attack can effectively shut down a smart building or factory.

Scenario A malicious actor floods a smart hospital's network with so much junk data that doctors cannot receive live updates from heart monitors.

Question (3-4 marks) Explain why IoT systems are specifically vulnerable to Denial of Service (DoS) attacks.

Mark Scheme:

- They have a heavy reliance on "communication channels" to function.
 - Many IoT devices have limited processing power and can be easily overwhelmed by traffic.
 - A disruption to the network can break the automation of critical tasks.
-

INSECURE COMMUNICATION CHANNELS

Definition Use of unencrypted public Wi-Fi or Bluetooth that allows eavesdropping.

Explanation Without encryption, the data being sent from a sensor to a hub can be intercepted and read by anyone nearby with the right equipment.

Scenario A smart lock transmits its "unlock" code over an unencrypted Bluetooth connection, allowing a nearby thief to capture the signal.

Question (3-4 marks) State a mitigation strategy for insecure communication channels in an IoT design.

Mark Scheme:

- Use of strong encryption protocols like SSL or TLS.
 - Ensuring all data is sent over encrypted/private networks rather than public Wi-Fi.
 - Implementing authentication to ensure only verified devices can join the channel.
-

SSL (SECURE SOCKET LAYER)

Definition A networking protocol providing secure connections.

Explanation It uses certificates and encryption to authenticate the identity of servers and ensure that data transmitted between devices remains private.

Scenario A smart home hub uses a protocol to ensure that the control commands sent from a smartphone app are encrypted and cannot be altered.

Question (3-4 marks) Describe the process of establishing a secure connection using SSL.

Mark Scheme:

- The browser/device requests server identification and the server sends an SSL certificate.
 - A trusted Certificate Authority (CA) is used to verify the certificate's authenticity.
 - A unique one-time symmetric session key is created to encrypt the data transmission.
-

TLS (TRANSPORT LAYER SECURITY)

Definition A secure version of the protocol used to transfer data... providing secure connections.

Explanation It is the modern, more secure successor to SSL, used to protect communication channels in IoT solutions from eavesdropping and tampering.

Scenario A bank's IoT network uses the latest transport security protocol to protect the transmission of ATM sensor data to the central office.

Question (3-4 marks) Explain the significance of TLS in protecting IoT data integrity.

Mark Scheme:

- Provides a "secure communication channel" between devices.
 - Uses encryption to prevent "unauthorized access" or eavesdropping during transmission.
 - Ensures that data has not been "tampered with" while traveling across the network.
-

IOT SOLUTION DESIGN

Definition Requires a written plan and an information flow diagram (IFD).

Explanation Designing a solution involves identifying the specific objectives, the necessary sensors/hardware, and how the information will flow between them to achieve automation.

Scenario An engineer creates a blueprint for an automated irrigation system that includes soil sensors, a central hub, and a connection to a weather API.

Question (3-4 marks) Identify three requirements that must be considered when designing an IoT solution.

Mark Scheme:

- Clear "objectives" (what the system is meant to achieve).
 - Identification of "individual components/sensors" needed.
 - Mapping of the "information flow" between devices and servers.
-

INFORMATION FLOW DIAGRAM (IFD)

Definition Visual representations showing how information travels within and between systems, organisations, and processes.

Explanation In IoT design, an IFD uses entities (sources), processes (transformations), and arrows (data flow) to map out how sensor data moves to hubs and users.

Scenario A diagram shows arrows moving from "Temperature Sensor" to a "Central Hub" box, which then points to a "User Smartphone" rectangle.

Question (3-4 marks) Explain the function of a "Decision" component (diamond shape) in an Information Flow Diagram.

Mark Scheme:

- Represents a logical decision point in the system.
 - Indicates that the flow of information depends on "specific conditions".
 - (Example: If temperature > 25, send alert; else, continue monitoring).
-

HOW TO ANSWER GUIDE

1. DEFINE (1-2 marks)

- **What to include:** Give the "Core Concept." Use the exact technical terminology (e.g., for Machine Learning, mention "without being explicitly programmed").
- **What to avoid:** Don't just repeat the word in the definition (e.g., "Virtual Reality is a reality that is virtual"). Avoid overly long sentences; keep it punchy.

2. EXPLAIN (3-4 marks)

- **Structure:** Use a **Point → Evidence → Explain** approach.
- **Keywords:** Use "leads to," "which results in," or "because."
- **Example:** "The test set is data the model hasn't seen (Point) because this allows the developer to evaluate the real-world accuracy of predictions (Explain), which ensures the model hasn't just memorized the training data (Result)."

3. DISCUSS / EVALUATE (6-8 marks)

- **Structure:** Provide a balanced argument (Pros vs. Cons or Method A vs. Method B).
- **Scenario Use:** You **MUST** refer back to the specific scenario in the question. If the question is about a smart hospital, your answer should mention patients, doctors, or medical sensors.
- **Examiner's Wants:** They want to see **AO3 skills:** a comparison of different storage types (Edge vs. Cloud) and a justified conclusion at the end based on the scenario's needs.

4. COMMON MISTAKES ON TOPIC 16

- **Confusing ML and Expert Systems:** Remember: Expert systems follow **fixed rules**; Machine Learning **identifies patterns** on its own.
- **Vague VR answers:** Don't just say "it looks real." Use technical terms like **stereoscopic screens, depth perception, and haptic feedback**.
- **IoT Security Overgeneralization:** Don't just say "it's unsecure." Specify **why** (default passwords, unpatched software, unencrypted channels) and **how** to fix it (SSL/TLS, encryption).
- **Misidentifying Learning Types:** Labelling data = Supervised. Finding hidden groups in raw data = Unsupervised.